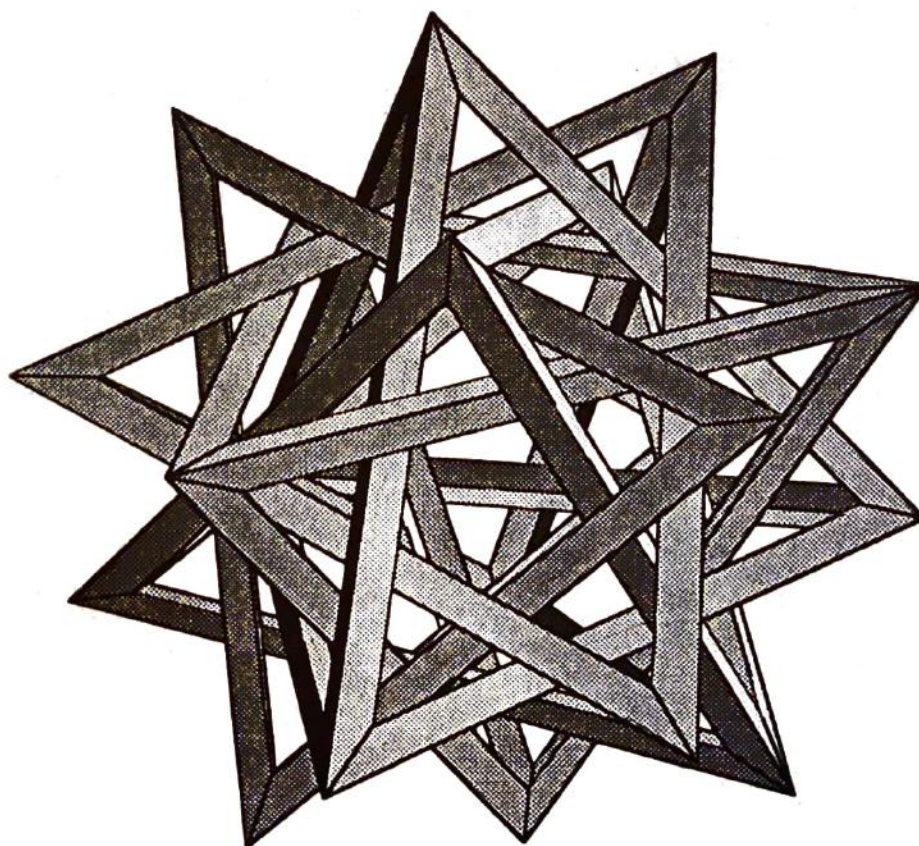


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From the Editor-in-Chief

This is the seventy-first volume in *Math Survey's* long history as the official mathematics publication of Stuyvesant High School. We are also continuing a somewhat shorter tradition (three years) of publishing two issues per year, a tradition that is difficult to maintain but is extremely rewarding, both for us, the staff, and for you, the reader.

As editor-in chief, I extend my sincerest thanks to our faculty advisor, Mr. Allen Clancy, for his efforts on our behalf; our department chairman, Dr. Richard Rothenberg, for his unstinting support; and our math team coach, Mr. Richard Geller, for his guidance. *Math Survey* also appreciates the efforts of all the teachers of the mathematics department for their input and assistance. I would like to thank Linda Hong for organizing the preparation of the publication as well as for meeting impossible deadlines; Nelson Uhan and John Choi for a hurried but excellent layout job; Edmund Chou and Tzyy-Ming Yeh for their advertising campaigns; and Daniel Stronger, Leo Nguyen, Vinci Liu, Felix Liu, Michael Shy, Abe Gurjal and Weikai Lang for their exhaustive proofreading and corrections. Nevertheless, there may still remain errors, for which I am solely responsible. Furthermore, *Math Survey* would never have been able to publish this issue without the timely aid of Mr. John Lapolla, Kieran Hervold, Brian Czyzewski, Jisoo Lee and the Student Union government. Last, I would like to thank you the reader, our generous advertisers and sponsors for supporting *Math Survey*.

We, the staff of *Math Survey*, hope that you will enjoy this issue.

Sincerely,

A handwritten signature in black ink, reading "Hyun-Sup Byun". The signature is written in a cursive, flowing style.

Hyun-Sup Byun
Editor-in-Chief

A New Approach to the Eigenvalue Problem

Linda Hong

1 Introduction

• 1a : The Main Results

Let A be an $n \times n$ matrix. The $n \times 1$ vector h , $h \neq 0$, is called an **eigenvector** of A if

$$Ah = \lambda h$$

for some scalar λ . This λ is called an **eigenvalue** of A and corresponds to the eigenvector h .

In the classical eigenvalue problem, the solutions for all the eigenvalues and their corresponding eigenvectors of any matrix can be obtained. However, when the matrix is large the solutions become difficult to find. Ideally, a computer program would do all the difficult computations for us, but one does not exist as of yet because of the nature of the solution method. In this approach, one of our objective goals will be to create a process of solution that can be used by a computer. Because eigenvalues are used in solving systems of differential equations, an easy method for calculating them would be very helpful in defining physical systems, such as systems of springs and systems of circuits.

Let us focus on positive definite symmetric matrices since it is known that their eigenvalues are positive and real. We will first consider the problem of evaluating the largest (senior) eigenvalue and its corresponding eigenvector in the case where there are simple characteristic roots. The proposed approach in a slightly modified form may also be used to evaluate the minimum (junior) eigenvalue and thereby generate the exact bounds of the spectrum of the matrix. Further investigations may lead to solving for the *full* eigenvalue problem for the set of all positive definite symmetric matrices with simple characteristic roots. The main features of the approach are:

- (i) reducing the problem to the unconstrained minimization problem for a multivariable potential function.
- (ii) solving the unconstrained minimization problem via ordinary differential equations.
- (iii) obtaining a numerical solution to ordinary differential equations in the CAUCHY program.
- (iv) describing the executed numerical experiments.

The main part will be the unconstrained variational description of the senior eigenvector in terms of the new potential function, Π . The problem of finding the senior pair can be reduced to the unconstrained minimization problem for a specially constructed function of n variables, $x = (x_1, \dots, x_n)$. This function, $\Pi(x)$, will be called the potential function associated with the matrix A , and will be described below. The *idea* of the potential function Π in the eigenvalue problem is closely connected with a similar idea used in ALPHA (See Reference [4]) on solving linear time-optimal control problems and similar control problems. Results of numerical experiments were obtained with the help of the CAUCHY program for solving initial value problems for ordinary differential equations. The CAUCHY program was created in Moscow State Lomonosov University, by the Department of Optimal Control (see Section 4). Note that CAUCHY was not intended by its creators to solve the eigenvalue problem.

• 1b : Definitions and Terminology

1) If the eigenvalues of A are $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ and their corresponding eigenvectors are $h_1, h_2, \dots, h_n$, then the **senior pair** of A is $\{\lambda_n, h_n\}$ and the **junior pair** is $\{\lambda_1, h_1\}$.

2) The **Kronecker delta** is represented by $\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$.

3) The identity matrix is the $n \times n$ matrix, $I = (\delta_{ij})$, $(i, j = 1, \dots, n)$, where the element in the i th row and j th column is δ_{ij} .

4) Matrix A is symmetric if when $A = (a_{ij})$, $a_{ij} = a_{ji}$, $(i, j = 1, \dots, n)$. In other words: $A^T = A$.

5) A set of vectors $\{v_1, v_2, \dots, v_n\}$ of unit norm is said to be orthonormal if they are pairwise orthogonal, or perpendicular. In other words: $v_i \bullet v_j = 0$, $i \neq j$, $\|v_i\| = 1$, $(i, j = 1, \dots, n)$.

6) If the norm of vector v is equal to unity ($\|v_i\| = 1$), then it is called a normed vector.

7) The quadratic form associated with matrix A is: $Q(x) = Ax \bullet x = x^T A x = \sum_{i,j=1}^n a_{ij} x_i x_j$, for column vector $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$.

8) Matrix A is positive definite if $x^T A x > 0$ for all real column vectors $x \neq 0$. A is negative definite if $x^T A x < 0$ or is indefinite if $x^T A x$ has both positive and negative values.

9) The characteristic equation of matrix A is of the following form: $\det(A - \lambda I) = 0$. Note that eigenvalues of matrix A are the roots of its characteristic equation. They can also be called the characteristic roots of A .

2 Theorems

All of the characteristic roots of a real symmetric matrix are real. All of the characteristic roots of a positive definite are positive. For the rest of this article, A will be a positive definite symmetric $n \times n$ matrix with distinct characteristic roots, or eigenvalues. Let us denote roots by: $\lambda_1, \lambda_2, \dots, \lambda_n$. We can say that $0 < \lambda_1 < \lambda_2 < \dots < \lambda_n$. There exists an orthonormal system, $h_1, h_2, \dots, h_n$ of eigenvectors of A such that:

$$A h_j = \lambda_j h_j, \quad j = 1, 2, \dots, n$$

Let's consider the quadratic form associated with matrix A :

$$Q(x) = Ax \bullet x = x^T A x \quad (1)$$

It is known that:

$$\lambda_n = \max_{\|x\|=1} (x^T A x) \quad (2)$$

$$\lambda_1 = \min_{\|x\|=1} (x^T A x) \quad (3)$$

Defining the positive definite quadratic form (1), we introduce the potential function Π as follows:

$$\Pi(x) = \frac{\|x\|^2 - \ln(x^T A x)}{2}, \quad x \neq 0$$

This function depends on n variables $x_1, x_2, \dots, x_n$, forming an n -dimensional vector, x .

Theorem 1: The senior normed eigenvector h_n , is a minimizing point of the function Π .

$$\min_{x \neq 0} \Pi(x) = \Pi(h_n) \quad \text{or} \quad h_n = \arg \min \Pi(x)$$

Vector $-h_n$ is also a minimizing point, and: $\Pi(x) > \Pi(\pm h_n) = \Pi_{\min} \quad \forall x, x \neq 0, \pm h_n$

PROOF: Theorem 1 follows from Theorem 4. $\square$

Theorem 2: The gradient, $\Pi'(x) \equiv \nabla \Pi(x)$, of the potential function $\Pi(x)$ may be presented in the following vector form

$$\Pi'(x) = x - \frac{Ax}{x^T A x}, \quad x \neq 0$$

and at points $x = \pm h_j$, $j = 1, 2, \dots, n$, we have: $\Pi'(\pm h_j) = 0$. Thus, all normed eigenvectors of A are critical points of Π .

PROOF: $((\|x\|^2)') = (x \bullet x)' = 2x$ and $(x^T A x)' = (Ax, x)' = 2Ax$.

So:

$$\nabla \Pi(\mathbf{x}) = \frac{(\|\mathbf{x}\|^2)'}{2} - \frac{(\mathbf{x}^T A \mathbf{x})'}{2 \mathbf{x}^T A \mathbf{x}} = \mathbf{x} - \frac{A \mathbf{x}}{\mathbf{x}^T A \mathbf{x}}$$

If $0 < \lambda_1 < \lambda_2 < \dots < \lambda_n$ are the eigenvalues of matrix A , and $\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n$ are their corresponding eigenvectors we know that for any i , $\mathbf{h}_i^T A \mathbf{h}_i = \lambda_i$ and $\mathbf{h}_i - \frac{\lambda_i \mathbf{h}_i}{\mathbf{h}_i^T A \mathbf{h}_i} = \mathbf{h}_i - \frac{A \mathbf{h}_i}{\lambda_i} = \mathbf{0}$. We can see that $\nabla \Pi(\mathbf{h}_i) = \mathbf{0}$, $i=1, \dots, n$.

Therefore the eigenvectors of A are the critical points of Π (except $\mathbf{x}=\mathbf{0}$). $\square$

Lemma: The $n \times n$ matrix A , with unit orthonormal eigenvectors $\{\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n\}$ and eigenvalues $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ can be expressed in the following form:

$$A = \lambda_1 \mathbf{h}_1 \mathbf{h}_1^T + \lambda_2 \mathbf{h}_2 \mathbf{h}_2^T + \dots + \lambda_n \mathbf{h}_n \mathbf{h}_n^T = \sum_{i=1}^n \lambda_i \mathbf{h}_i \mathbf{h}_i^T$$

This is called the spectral form of A . It follows that

$$I = \mathbf{h}_1 \mathbf{h}_1^T + \mathbf{h}_2 \mathbf{h}_2^T + \dots + \mathbf{h}_n \mathbf{h}_n^T = \sum_{i=1}^n \mathbf{h}_i \mathbf{h}_i^T$$

Theorem 3: The Hessian $\Pi''(\mathbf{x})$ can be represented in the following way:

$$\Pi''(\mathbf{x}) = I - \frac{A \mathbf{x}}{\mathbf{x}^T A \mathbf{x}} + 2 \frac{[A \mathbf{x}][A \mathbf{x}]^T}{(\mathbf{x}^T A \mathbf{x})^2}$$

and at the points $\mathbf{x} = \pm \mathbf{h}_n$ it is positive definite. However, where $\mathbf{x}$ is equal to the other normed eigenvectors $\{\pm \mathbf{h}_1, \dots, \pm \mathbf{h}_{n-1}\}$, $\Pi''(\mathbf{x})$ will be indefinite.

PROOF: We know that $\nabla \Pi(\mathbf{x}) = \mathbf{x} - \frac{A \mathbf{x}}{\mathbf{x}^T A \mathbf{x}}$.

So:

$$\Pi''(\mathbf{x}) = (\mathbf{x})' - \frac{(A \mathbf{x})'}{\mathbf{x}^T A \mathbf{x}} + \frac{(A \mathbf{x})(\mathbf{x}^T A \mathbf{x})'}{(\mathbf{x}^T A \mathbf{x})^2} = I - \frac{A}{\mathbf{x}^T A \mathbf{x}} + 2 \frac{[A \mathbf{x}][A \mathbf{x}]^T}{(\mathbf{x}^T A \mathbf{x})^2}$$

When $\mathbf{x} = \mathbf{h}_k$ for some k :

$$\Pi''(\mathbf{h}_k) = I - \frac{A}{\lambda_k} + \frac{2 \lambda_k^2 \mathbf{h}_k \mathbf{h}_k^T}{\lambda_k^2}$$

Then from Lemma 3.1 let us express A and I in their spectral forms.

$$\Pi''(\mathbf{h}_k) = \sum_{i=1}^n \mathbf{h}_i \mathbf{h}_i^T - \frac{1}{\lambda_k} \sum_{i=1}^n \lambda_i \mathbf{h}_i \mathbf{h}_i^T + 2 \mathbf{h}_k \mathbf{h}_k^T = \sum_{i=1}^n (1 - \frac{\lambda_i}{\lambda_k}) \mathbf{h}_i \mathbf{h}_i^T + 2 \mathbf{h}_k \mathbf{h}_k^T$$

Or we can say that: $\Pi''(\mathbf{h}_k) = \sum_{i=1}^n \mu_i \mathbf{h}_i \mathbf{h}_i^T$, which is the spectral form of $\Pi''(\mathbf{h}_k)$ with the $\mathbf{h}_i$'s as its eigenvectors and the μ_i 's as its eigenvalues. Let us expand the spectral for any $k < n$:

$$\Pi''(\mathbf{h}_k) = (1 - \frac{\lambda_1}{\lambda_k}) \mathbf{h}_1 \mathbf{h}_1^T + \dots + 2 \mathbf{h}_k \mathbf{h}_k^T + (1 - \frac{\lambda_{k+1}}{\lambda_k}) \mathbf{h}_{k+1} \mathbf{h}_{k+1}^T + \dots + (1 - \frac{\lambda_n}{\lambda_k}) \mathbf{h}_n \mathbf{h}_n^T$$

We can clearly see that for all $i < k$, $\mu_i > 0$ and for all $i \geq k$, $\mu_i < 0$. Thus the matrix is indefinite. For $\mathbf{h}_n$:

$$\Pi''(\mathbf{h}_n) = (1 - \frac{\lambda_1}{\lambda_n}) \mathbf{h}_1 \mathbf{h}_1^T + \dots + (1 - \frac{\lambda_{n-1}}{\lambda_n}) \mathbf{h}_{n-1} \mathbf{h}_{n-1}^T + 2 \mathbf{h}_n \mathbf{h}_n^T$$

and all the μ_i 's, or eigenvalues will be positive. Therefore $\Pi''(\mathbf{h}_k)$ is positive definite. $\square$

Theorem 4: $\pm \mathbf{h}_n$ are the minimum points of $\Pi(\mathbf{x})$; $\{\pm \mathbf{h}_1, \dots, \pm \mathbf{h}_{n-1}\}$ are saddle points of $\Pi(\mathbf{x})$.

PROOF: From Theorem 3, we can immediately conclude that $\pm \mathbf{h}_n$ are relative minimum points and that $\{\pm \mathbf{h}_1, \dots, \pm \mathbf{h}_{n-1}\}$ are saddle points of the function $\Pi(\mathbf{x})$. All that needs to be shown is that $\Pi(\mathbf{x})$ is not minimized when $\mathbf{x}$ is zero or infinity, i.e.-we wish to show that $\lim_{\|\mathbf{x}\| \rightarrow \infty, 0} \Pi(\mathbf{x}) = +\infty$.

It is known that: $\lambda_1 \mathbf{x}^T \mathbf{x} \leq \mathbf{x}^T A \mathbf{x} \leq \lambda_n \mathbf{x}^T \mathbf{x}$ (See equations (2) and (3)). So:

$$\frac{\|\mathbf{x}\|^2 - \ln(\mathbf{x}^T A \mathbf{x})}{2} \geq \frac{\|\mathbf{x}\|^2 - \ln(\lambda_n \|\mathbf{x}\|^2)}{2}$$

When we take the limit as $\|\mathbf{x}\|$ approaches zero or infinity, the right side of the inequality approaches infinity. The left side, $\Pi(\mathbf{x})$, also approaches infinity which is definitely greater than $\Pi(\pm \mathbf{h}_n)$. Therefore $\pm \mathbf{h}_n$ are the minimum points of Π . $\square$

Another simple way to verify that $\pm \mathbf{h}_n$ are the minimum points is to find the values of $\Pi(\mathbf{x})$ when $\mathbf{x}$ is a critical point, or an eigenvector:

$$\Pi(\mathbf{h}_i) = \frac{\|\mathbf{h}_i\|^2 - \ln(\mathbf{h}_i^T A \mathbf{h}_i)}{2} = \frac{1 - \ln(\lambda_i)}{2} \Rightarrow \frac{1 - \ln(\lambda_1)}{2} > \frac{1 - \ln(\lambda_2)}{2} > \dots > \frac{1 - \ln(\lambda_n)}{2}$$

As you can see $\Pi(\mathbf{h}_n)$ is the least out of all of these, and $\pm \mathbf{h}_n$ must be the minimum points. In other words: $\Pi(\mathbf{x}) > \Pi(\pm \mathbf{h}_n) = \Pi_{\min}$, $\forall \mathbf{x}$, $\mathbf{x} \neq 0, \pm \mathbf{h}_n$. Note that Theorem 1 has also been proven here.

For Theorem 5, we will need the following. Let us consider the nonlinear vectorial differential equation:

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}), f(\mathbf{a}) = 0, \mathbf{x} = (x_1, \dots, x_n), f(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_n(\mathbf{x})) \quad (4)$$

with equilibrium point $\mathbf{x} = \mathbf{a}$ and function $f(\mathbf{x})$ smooth enough in some neighborhood of point $\mathbf{a}$. The $n \times n$ matrix

$$A \text{ is defined by: } A = f'(\mathbf{a}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{a}) & \dots & \frac{\partial f_1}{\partial x_n}(\mathbf{a}) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1}(\mathbf{a}) & \dots & \frac{\partial f_n}{\partial x_n}(\mathbf{a}) \end{pmatrix}$$

The linear system $\frac{d\mathbf{x}}{dt} = A\mathbf{x}$ is called the system of first approximation for the nonlinear system (4) at rest point

$\mathbf{a}$.

Lyapunov's Theorem: If all eigenvalues of A have negative real parts then $\mathbf{a}$ is an *asymptotically stable equilibrium point* of the nonlinear equation (4). (See References [2]). $\square$

Theorem 5: $\pm \mathbf{h}_n$ are stable equilibrium (rest) points of the vector gradient descent differential equation:

$$\frac{d\mathbf{x}}{dt} = -\Pi'(\mathbf{x}) \quad (5)$$

where $\Pi'(\mathbf{x})$ is the gradient of the potential function Π . $\{\pm \mathbf{h}_1, \dots, \pm \mathbf{h}_{n-1}\}$ are unstable equilibrium (rest or saddle-type) points of equation (5).

PROOF: Using Lyapunov's theorem of stabilization, all we need prove is that all the eigenvalues of $-\Pi''(\pm \mathbf{h}_n)$ have negative real parts. From the proof of Theorem 3 we have:

$$\Pi''(\pm \mathbf{h}_n) = \sum_{i=1}^n \mu_i \mathbf{h}_i \mathbf{h}_i^T$$

with $\mu_i > 0$, $i = 1, \dots, n$. Consequently, the eigenvalues of $-\Pi''(\pm \mathbf{h}_n)$ are negative. Therefore, $\pm \mathbf{h}_n$ are the stabilization points of the trajectory described by $\dot{\mathbf{x}} = -\nabla \Pi(\mathbf{x})$.

In addition, there exists at least one positive and one negative eigenvalue of $-\Pi''$ for any other eigenvector. Thus all other equilibrium points of the trajectory will be unstable. $\square$

Theorem 5 forms the basis for the computational procedures that will calculate the senior eigenvector, $\mathbf{h}_n$, via the potential function Π . $\mathbf{h}_n$ can be found by solving the unconstrained minimization problem

$$\Pi(\mathbf{x}) \rightarrow \min_{\mathbf{x} \neq \mathbf{0}}$$

which can be numerically computed using the CAUCHY program.

The object of the minimization procedure is to search for the stable equilibrium point of the ordinary vectorial differential equation of gradient descent (5). For computational convenience, a special variation of the equation will be used:

$$\dot{\mathbf{x}} = -k \frac{\Pi'(\mathbf{x})}{\sqrt{1 + \|\Pi'(\mathbf{x})\|^2}} \quad (6)$$

where the positive multiplier k is chosen by the user and $\dot{\mathbf{x}}$ is the derivative of $\mathbf{x}$ with respect to time. The multiplier, k , is introduced to regulate the velocity along the trajectories of system (6). In (6) the norm of $\dot{\mathbf{x}}$ will be approximately equal to k . Note that this system and the regular system have common equilibrium points and common trajectories in n -space.

We can choose the initial point $\mathbf{x}_0 \neq \mathbf{0}$ and introduce the initial condition

$$\mathbf{x}(t)|_{t=0} = \mathbf{x}_0 \quad (7)$$

The problem is now simply the initial value (Cauchy) problem (6), (7). The solution, $\mathbf{x}(t, \mathbf{x}_0)$, to the problem has the important stabilization property:

$$\lim_{t \rightarrow +\infty} \mathbf{x}(t, \mathbf{x}_0) = \mathbf{h}_n \quad (8)$$

or, depending on $\mathbf{x}_0$:

$$\lim_{t \rightarrow +\infty} \mathbf{x}(t, \mathbf{x}_0) = -\mathbf{h}_n \quad (9)$$

In addition, the point $\mathbf{x}_0$ should not coincide with the normed eigenvectors

$$\pm \mathbf{h}_1, \dots, \pm \mathbf{h}_{n-1}, \quad (\|\mathbf{h}_i\| = 1) \quad (10)$$

which are unstable equilibrium points of the trajectories described by (6) and the original equation.

The evaluation of the senior eigenvalue is based on the limit relations (8) and (9) via solving the initial value problem (6), (7) on some finite interval of time $[0, T]$, $0 \leq t \leq T$, using CAUCHY. The parameters T (our hypothetical infinity) and k can be chosen randomly by the user and one may observe the development of the trajectory on the computer screen. We will see that as t approaches T , the trajectory will stabilize at one of the points $\pm \mathbf{h}_n$. After finding the senior eigenvector, direct calculation gives the corresponding eigenvalue: $\lambda_n = \mathbf{h}_n^T A \mathbf{h}_n$. There is a short description of the CAUCHY program in Section 4.

3 Explanation of the approach using a simple example

Before we go on it will be necessary to introduce the following theorem:

Sylvester's Theorem: The symmetric $n \times n$ matrix A is a positive definite matrix *iff* when

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \quad \text{and} \quad \begin{cases} \Delta_1 = |a_{11}| \\ \Delta_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \\ \dots\dots\dots \\ \Delta_n = \det[A] \end{cases} \quad \text{the following is true:} \quad \begin{cases} \Delta_1 > 0 \\ \Delta_2 > 0 \\ \dots\dots\dots \\ \Delta_n > 0 \end{cases}.$$

A is negative definite if the signs of the Δ 's alternate between negative and positive starting with $\Delta_1 < 0$. $\square$

If the Hessian matrix or second derivative of a function at a critical point is positive definite, then the point is a relative minimum of the function. If the result is instead negative definite, then the point is a relative maximum.

Example: Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$.

•We can find the eigenvectors of A in the usual manner as follows:

We know that the eigenvalues are the roots of the characteristic equation $\det(A - \lambda I) = 0$. So:

$$\begin{vmatrix} 1-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} = 0 \Rightarrow (1-\lambda)(2-\lambda) = 0 \text{ AND } \lambda_1=1, \lambda_2=2$$

We can then solve for the normed eigenvectors, $\mathbf{h}_1$ and $\mathbf{h}_2$ using the equation: $\begin{pmatrix} 1-\lambda & 0 \\ 0 & 2-\lambda \end{pmatrix} \mathbf{h} = \mathbf{0}$. Solving for each component of each vector, we get the following: $\mathbf{h}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{h}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

•Using the Π function we have:

$$\Pi(\mathbf{x}) = \frac{x_1^2 + x_2^2 - \ln(x_1^2 + 2x_2^2)}{2} \quad \text{and} \quad \nabla \Pi(\mathbf{x}) = \begin{pmatrix} x_1 - \frac{x_1}{x_1^2 + 2x_2^2} \\ x_2 - \frac{2x_2}{x_1^2 + 2x_2^2} \end{pmatrix}$$

Setting gradient to zero, we find the critical points are $\pm \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\pm \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ with $\lambda_1 = 1$ and $\lambda_2 = 2$, which are the same as in the previous method.

We can also verify that $\pm \mathbf{h}_2$ are the absolute minimum points of Π by finding the function's Hessian:

$$\Pi''(\mathbf{x}) = \begin{pmatrix} 1 + \frac{x_1^2 - 2x_2^2}{(x_1^2 + 2x_2^2)^2} & \frac{4x_1x_2}{(x_1^2 + 2x_2^2)^2} \\ \frac{4x_1x_2}{(x_1^2 + 2x_2^2)^2} & 1 + \frac{4x_2^2 - 2x_1^2}{(x_1^2 + 2x_2^2)^2} \end{pmatrix}$$

At points $\pm \mathbf{h}_1 = \pm \begin{pmatrix} 1 \\ 0 \end{pmatrix}$: $\Pi''(\pm \mathbf{h}_1) = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$, with $\Delta_1 = 2 > 0$ and $\Delta_2 = -2 < 0$. By Sylvester's Theorem, $\pm \mathbf{h}_1$

are saddle-points. For $\pm \mathbf{h}_2 = \pm \begin{pmatrix} 0 \\ 1 \end{pmatrix}$: $\Pi''(\mathbf{h}_2) = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix}$, and $\Delta_1 = 1/2 > 0$ and $\Delta_2 = 1 > 0$, which makes $\pm \mathbf{h}_2$ minimum points.

Moreover, these minimum points are the *absolute* minimum points of Π because at zero and infinity, Π approaches positive infinity.

The function Π is more efficient than the usual method in one manner because the eigenvectors are found before the eigenvalues rather than vice versa. We will now illustrate the procedure using the CAUCHY program to approximate the normed senior eigenvector for this simple example. The eigenvalues can be obtained easily after and we will see that the problem is considerably easier to solve this way. Although the numbers will be approximations, they will be good enough for most practical situations.

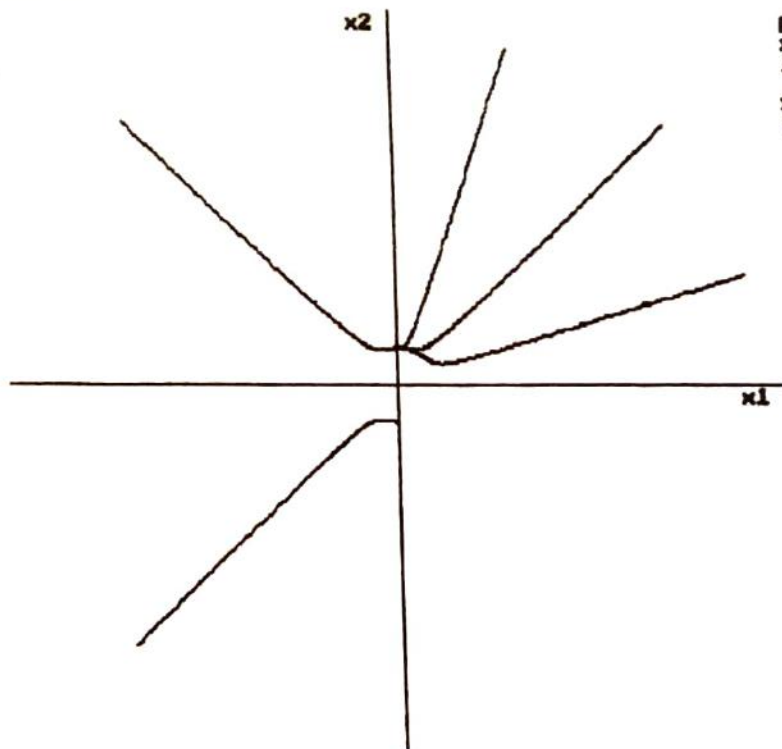
Using Theorem 5, if $\frac{d\mathbf{x}}{dt} = -\nabla \Pi(\mathbf{x})$ then as t approaches infinity, $\mathbf{x}$ will approach the normed senior eigenvector of A . So we can enter the following equations into the CAUCHY program:

$$\begin{cases} x_1' = -x_1 + x_1 / (x_1^2 + 2 * x_2^2) \\ x_2' = -x_2 + 2 * x_2 / (x_1^2 + 2 * x_2^2) \end{cases}$$

On the CAUCHY program we can graph up to 5 different trajectories each with different initial points. The range for time, t , should be large enough so that we can see the behavior of the trajectory as time approaches "infinity". We can also choose any of the four methods listed in Section 4 or change the number of steps in the process, to improve the accuracy.

In this experiment, the initial points $(7, 7)$, $(-7, 7)$, $(-7, -7)$, $(3, 9)$, $(9, 3)$ were chosen for the five processes, or trajectories. The range for t was from 0 to 20 and there were 1000 output points. As you can see in the graph below, all the trajectories converged on either $(0, 1)$ or $(0, -1)$. We can check the values of the coordinates in BROWSE - RESULTS. For the initial point $(7, 7)$, at $t = 20$, we find that $x_1 = 0.00012011692358$ which is extremely close to 0 and $x_2 = .99999999279$ which is very close to 1. (Table 1 contains the results for the initial point $(7, 7)$). If we wish to have a better approximation, we may pick a larger upper limit for t .

Min:
x1
-10
x2
-10



Max:
x1
10
x2
10

Table 1:

t	1.99999999998E+01
x1	1.2022682358E-04
x2	9.9999999279E-01

Figure 1

We can also use the modified version of this equation with the multiplier, k (See equation (6)). So here we would enter:

$$\dot{\mathbf{x}} = -k \frac{\Pi'(\mathbf{x})}{\sqrt{1 + \|\Pi'(\mathbf{x})\|^2}}$$

or:

$$\begin{cases} f_1 = x_1^2 + 2 * x_2^2 & f_5 = k * f_2 / f_4 \\ f_2 = -x_1 + x_1 / f_1 & f_6 = k * f_3 / f_4 \\ f_3 = -x_2 + 2 * x_2 / f_1 & x_1' = f_5 \\ f_4 = \text{sqrt}(1 + f_2^2 + f_3^2) & x_2' = f_6 \end{cases}$$

(f_1, f_2, f_3, f_4, f_5 , and f_6 are for our convenience and are not graphed in CAUCHY). Here we can control the velocity of the trajectory with the multiplier k . For this experiment: $t = 0$ to 10, $\mathbf{x}_0 = (7, 7)$, and there were 2000

output points. For Figure 2 and Table 2 we chose $k=1$; in Figure 3 and Table 3 we chose $k=10$. As you can see, the trajectory curve is steeper and reaches the equilibrium point at a greater rate when $k=10$; at $k=1$ the final point is $(0.51008656951, 0.94357470479)$ where as when $k=10$ the coordinates are $(0.00000000000, 0.99999999998)$, with 10 decimal point accuracy. The k multiplier makes it easier to modify the output.

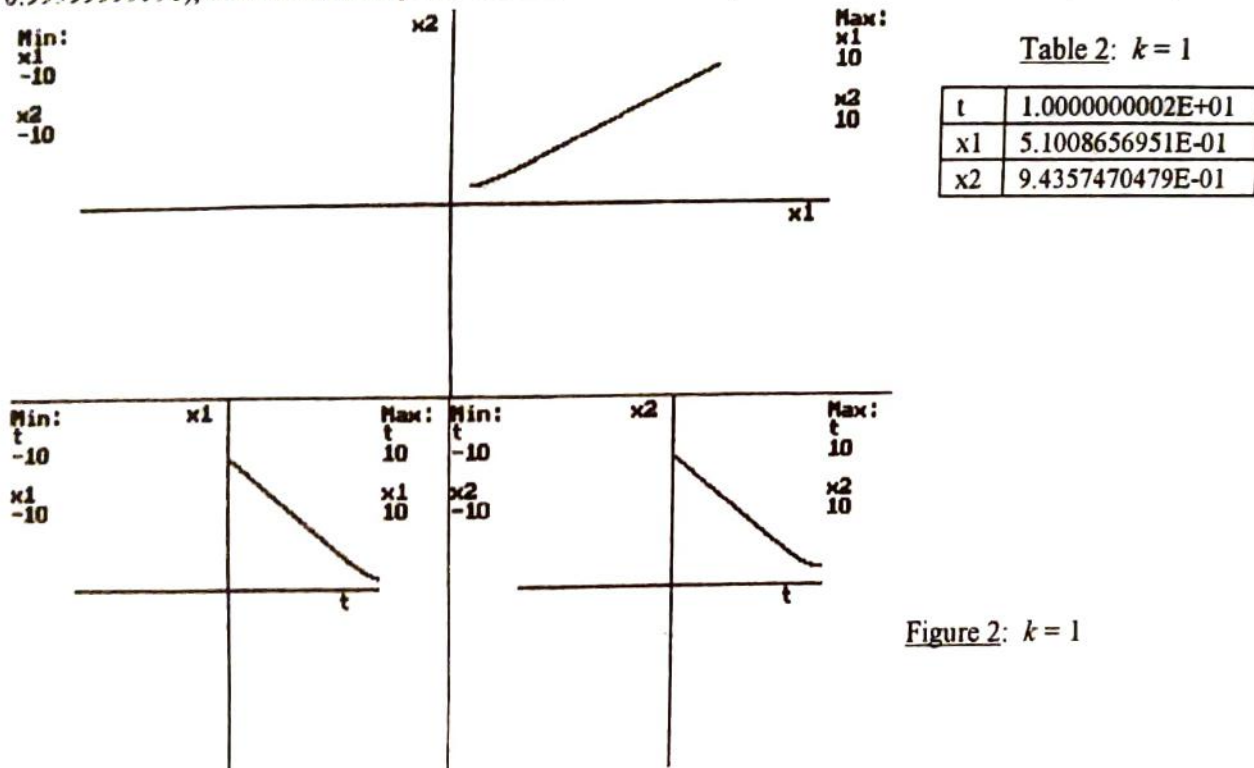


Figure 2: $k = 1$

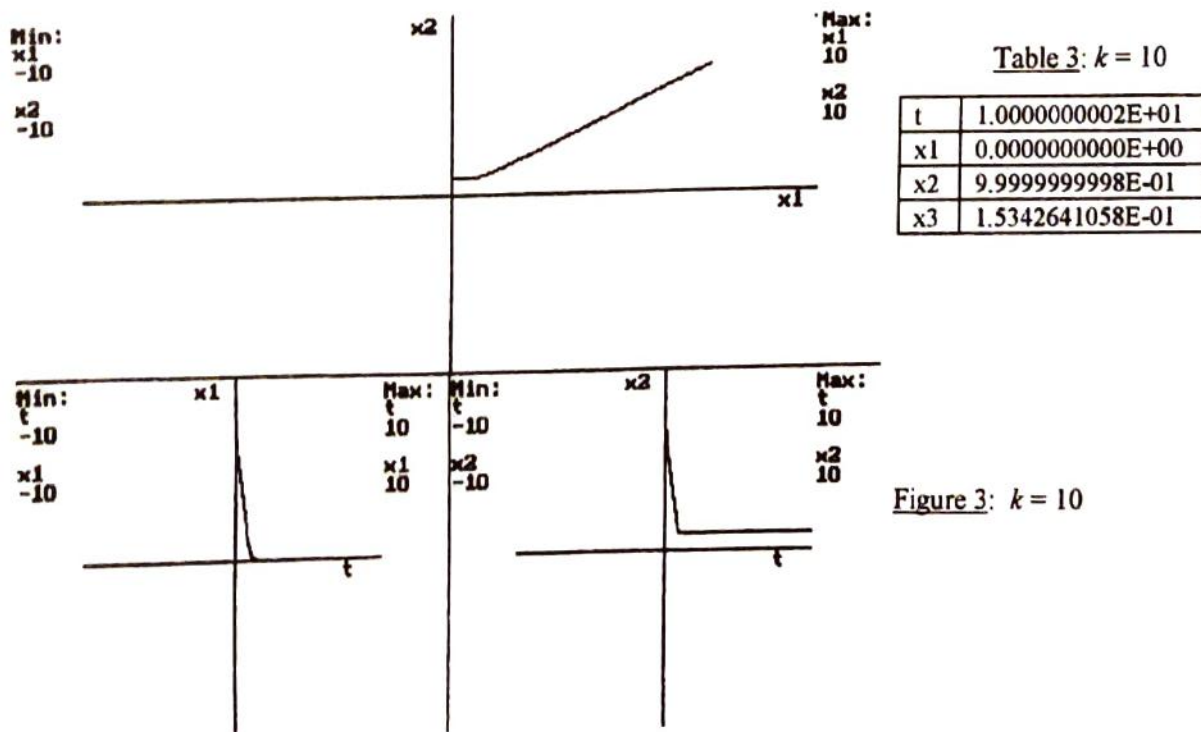


Figure 3: $k = 10$

The senior eigenvalue, λ_2 , can also be calculated using CAUCHY. Since we know that $\Pi(h_n) = \frac{1 - \ln(\lambda_n)}{2}$, we can facilitate our process by letting another variable, x_3 , equal to $\Pi(x)$. Using the chain rule,

$$\dot{x}_3 = \Pi'(x)_{x1} \dot{x}_1 + \Pi'(x)_{x2} \dot{x}_2$$

Since we found that $\Pi'(x)_{x1} = -f2$, $\Pi'(x)_{x2} = -f3$, $\dot{x}_1 = f5$, and $\dot{x}_2 = f6$, we can enter the following into CAUCHY:

$$x3' = -f2 * f5 - f3 * f6$$

The initial value of x_3 is $\Pi(x_0)$ which we calculate to be 46.50478371. At the upper limit of t , $\lambda_2 = e^{1-2x_3}$. The results of this experiment can be seen in Table 3, where $k = 10$. $x_3 = 0.15342641058$ and we calculate that $\lambda_2 \approx 1.999999997$, which is very close to the true value of 2.

4 About CAUCHY

The CAUCHY program is designed so that the user can obtain a numerical solution to the Cauchy problem for a system of ordinary differential equations:

$$\mathbf{x}' = \mathbf{F}(\mathbf{x}, t), \quad \mathbf{x}(t_0) = \mathbf{x}_0, \quad t_0 \leq t \leq t_1$$

where $\mathbf{x} = \mathbf{x}(t)$ is an n -dimensional vector, t is an independent variable, $\mathbf{F}$ is an arbitrary function of variables $\mathbf{x}$ and t , and $\mathbf{x}_0$ is the initial condition. The maximum order of the system is 15. Graphs are plotted using results from calculation.

The system of equations can be solved by any one of the following predefined numerical methods:

- Euler method
- Runge-Kutta 2nd and 4th order methods
- Runge-Kutta-Fehlberg automatic step control method.

In this program, one can determine up to five different sets of phase variable values simultaneously for a single system of equations. Each graph is displayed in a different color, or with different line styles on monochrome displays with a maximum of three processes. Equations are entered into the built-in screen editor and can have up to 15 parameters named by a single letter (except t). The CAUCHY program graphs the resulting points on up to four separate windows. The user can change the minimum and maximum axes values, the axes labels, or the dimension between 2 and 3 dimensions. The user can also zoom in and out, move the center of the axes, or rotate the axes in the three dimensional window. There are some built in functions (sin, cos, ln, etc.) and a built-in calculator.

5 Conclusion

A solution to the eigenvalue problem for positive definite symmetric matrices has been successfully created that does not involve as many complex calculations as the classical method. Since we have solved for the senior pair, we can now find a method for solving the full eigenvalue problem for this class of matrices. Let us begin with the junior (least) eigenvector-eigenvalue pair, $\{h_1, \lambda_1\}$

If we can find a way to express the given positive definite matrix A so that the junior eigenvalue can be solved for in a similar manner to the senior eigenvector, there is no need to go further. For example, if we use the matrix $-A$ instead of A , we find that the junior eigenvector of A will become the senior eigenvector of $-A$. Since $-A$ is necessarily a negative definite matrix, let us use a matrix of the following form

$$B = -A + \kappa I, \quad \kappa > 0$$

with κ large enough so that B will be a positive definite, i.e. $\kappa > \lambda_n$. If we write B in its spectral form we find that the eigenvalues will be $\kappa - \lambda_i$. The senior eigenvalue of B will be $\kappa - \lambda_1$, which can be found using the proposed method.

$$B = \sum_{i=1}^n \kappa h_i h_i^T - \sum_{i=1}^n \lambda_i h_i h_i^T = \sum_{i=1}^n (\kappa - \lambda_i) h_i h_i^T$$

With the exact bounds of the spectrum of the eigenvalues of A , the full eigenvalue problem can be solved. For example we can solve for λ_{n-1} by using the following equation:

$$C = A - \mu h_n h_n^T = \sum_{i=1}^{n-1} \lambda_i h_i h_i^T + (\lambda_n - \mu) h_n h_n^T$$

where $0 < \lambda_n - \mu < \lambda_1$ (Note that $\lambda_n - \mu$ is also the new junior eigenvalue). We can find λ_{n-1} by using the new method and then solve for all subsequent eigenvalues in a similar manner. Therefore, full eigenvalue problem is solved for the set of positive definite symmetric matrices.

In this article we have seen a new approach to the classical eigenvalue problem for positive definite symmetric matrices with simple characteristic roots. To find the senior eigenvector and corresponding eigenvalue, the potential function Π , associated with the matrix, was introduced so that the senior eigenvector was the minimizing point of Π . Practical calculations for finding the unconstrained minimum point of the nonlinear potential function Π were done on the basis of a modified differential equation of steepest descent in the CAUCHY program illustrated by a few examples. The eigenvalue problem for positive definite symmetric matrices was solved: it was reduced to several problems of finding the senior pair. The results from this project may be considered as the basis for creating programs, independent of CAUCHY, to solve the eigenvalue problem for the above class of matrices, not restricted to 15 dimensions.

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Fibonacci Numbers in Prime Moduli

Ping-Hsin Lee

The Fibonacci sequence was introduced by Leonardo Fibonacci of Pisa through the following problem: *A man bought a pair of immortal rabbits and bred them. After each month, the rabbits bred one pair of young. Each pair of young rabbits took one month to mature and then began to reproduce one pair of rabbits every month thereafter. What is the total numbers of rabbit pairs that the man has each month?*

The answer to this problem is a series of numbers, starting from the first month: 1, 2, 3, 5, 8, This came to be called the Fibonacci sequence, named after the mathematician that proposed this problem. It can be defined mathematically as: $F_1 = 1$, $F_2 = 1$, and $F_n = F_{n-2} + F_{n-1}$ for $n > 2$.

In this article I will investigate the behavior of Fibonacci numbers in prime moduli. In particular, the following conjecture will be proven:

Conjecture. For prime p , a) $F_{p-1} \equiv 0 \pmod p$ if $p \equiv \pm 1 \pmod 5$
b) $F_{p+1} \equiv 0 \pmod p$ if $p \equiv \pm 2 \pmod 5$.

First, one well known fact about Fibonacci numbers is that they can be generated using roots of a quadratic equation. The equation $k^2 - k - 1 = 0$ has two roots, $x = \frac{1+\sqrt{5}}{2}$ and $y = -\frac{1}{x} = \frac{1-\sqrt{5}}{2}$. The n th Fibonacci number can be expressed as $\frac{x^n - y^n}{x - y}$, or $\left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n \right]$. I shall proceed to state my own proof of this theorem.

Lemma 1. $F_n = \left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n \right]$

Proof. Proof by induction.

1. For $n = 1$, $F_1 = \left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right) - \left(\frac{1-\sqrt{5}}{2}\right) \right] = 1$. For $n = 2$, $F_2 = \left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^2 - \left(\frac{1-\sqrt{5}}{2}\right)^2 \right] = 1$.

2. Assume statement is true for $n = k-1$ and $n = k$.

$$F_{k-1} = \left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1} - \left(\frac{1-\sqrt{5}}{2}\right)^{k-1} \right] \text{ and } F_k = \left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^k - \left(\frac{1-\sqrt{5}}{2}\right)^k \right]$$

3. Prove $n = k+1$ is true

$$\begin{aligned} F_{k+1} &= F_{k-1} + F_k = \left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1} - \left(\frac{1-\sqrt{5}}{2}\right)^{k-1} \right] + \left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^k - \left(\frac{1-\sqrt{5}}{2}\right)^k \right] \\ &= \left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1} \left(1 + \frac{1+\sqrt{5}}{2}\right) - \left(\frac{1-\sqrt{5}}{2}\right)^{k-1} \left(1 + \frac{1-\sqrt{5}}{2}\right) \right] \end{aligned}$$

$$= \left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1} \left(\frac{1+\sqrt{5}}{2}\right)^2 - \left(\frac{1-\sqrt{5}}{2}\right)^{k-1} \left(\frac{1-\sqrt{5}}{2}\right)^2 \right]$$

$$= \left(\frac{1}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^{k+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{k+1} \right]. \text{ QED.}$$

Another sequence that is closely related with the Fibonacci sequence is the Lucas sequence. The Lucas sequence is defined as follows: $L_1 = 1$, $L_2 = 3$, and $L_n = L_{n-2} + L_{n-1}$. Lucas numbers can be generated by the same

quadratic equation $k^2 - k - 1 = 0$ with two roots, $x = \frac{1+\sqrt{5}}{2}$ and $y = -\frac{1}{x} = \frac{1-\sqrt{5}}{2}$. The n^{th} term of Lucas sequence can

be expressed as $x^n + y^n$ or $\left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$, a formula which can also be proven using induction. The two

sequences also share many important properties, including the two well-known theorems below:

Lemma 2.

- i. $2F_{m+n} = F_m L_n + F_n L_m$
- ii. $(L_n)^2 - 5(F_n)^2 = 4(-1)^n$
- iii. $(F_n)^2 - F_{n-1} F_{n+1} = (-1)^{n-1}$
- iv. $(L_n)^2 - L_{n-1} L_{n+1} = 5(-1)^n$

These proofs are relatively straightforward and will not be given here. However, the proofs for Lemma 3 are not so obvious.

Lemma 3.

- i. $(F_n, F_{n+1}) = 1$, $(L_n, L_{n+1}) = 1$, where for $(m, n) = d$, d is the greatest common divisor of m and n .
- ii. L_n and F_n are both odd or both even, and $(L_n, F_n) = 1$, $(L_n, F_n) = 2$ respectively.
- iii. $F_n | F_{kn}$ for all positive integers k .
- iv. If $(m, n) = d$, then $(F_m, F_n) = F_d$.

Proof.

- i. Let $(F_n, F_{n+1}) = d$. Then $d | F_n$ and $d | F_{n+1}$. Since $F_{n-1} = F_{n+1} - F_n$, $d | F_{n-1}$. Repeating this process, we can see that $F_{n-2}, F_{n-3}, \dots$ and so on are all divisible by d . Following an infinite descent, we can say that $d | F_1$. But $F_1 = 1$, so $d = 1$. Similarly, if $(L_n, L_{n+1}) = d$, then necessarily $d = 1$.
- ii. From Lemma 2 part (ii) we know that $(L_n)^2 - 5(F_n)^2 = 4$ or -4 . Consider the parity of the left side, $(L_n)^2 - 5(F_n)^2$. The expression is even if and only if L_n and F_n are of the same parity.

Case 1: L_n and F_n are both odd.

Let $(L_n, F_n) = d$. Since $d | F_n$ and $d | L_n$, d must be odd. Also, we see that $d | (L_n)^2 - 5(F_n)^2 = 4$. But the only positive divisors of 4 are 1, 2, and 4, so $d = 1$.

Case 2: L_n and F_n are both even.

Let $(L_n, F_n) = d$. Since L_n and F_n are both even, L_n^2 and F_n^2 are both divisible by 4. Given the

equation in Lemma 2 part (ii), $\left(\frac{L_n}{2}\right)^2 - 5\left(\frac{F_n}{2}\right)^2 = \pm 1$. Let $(L_n/2, F_n/2) = k$. $k | (L_n/2)$ and $k | (F_n/2)$,

so $k | \left(\frac{L_n}{2}\right)^2 - 5\left(\frac{F_n}{2}\right)^2 = \pm 1$. Therefore $k = 1$. So $(L_n, F_n) = d = 2$.

iii. We will use strong induction to prove this.

1. For $k = 1$, $F_n | F_n$, and the claim is true.
2. Assume that the claim is true for $k = 1, 2, \dots, r-1$.
So, $F_n | F_{kn}$ for $k = 1, 2, \dots, r-1$.
3. Prove the claim true for r .

From Lemma 2 part (i), $2F_r = F_n L_{(r-1)n} + F_{(r-1)n} L_n$

Case 1: F_n is odd. By assumption, $F_n | F_{(r-1)n}$. Since $F_n | [F_n L_{(r-1)n} + F_{(r-1)n} L_n]$, $F_n | 2F_r$. Since F_n is odd, $F_n | F_r$.

Case 2: F_n is even. Then by Lemma 3 part (ii) L_n is even. Furthermore, $F_{(r-1)n}$ and $L_{(r-1)n}$ are even since $F_n | F_{(r-1)n}$ and $L_n | L_{(r-1)n}$ by assumption. So we can divide both sides of the equation by 2 and obtain $F_r = F_n (\frac{1}{2} L_{(r-1)n}) + F_{(r-1)n} (\frac{1}{2} L_n)$, where $\frac{1}{2} L_{(r-1)n}$ and $\frac{1}{2} L_n$ are integers. Again, by assumption $F_n | F_{(r-1)n}$ and thus $F_n | [F_n (\frac{1}{2} L_{(r-1)n}) + F_{(r-1)n} (\frac{1}{2} L_n)] = F_r$.

4. If $(m, n) = d$, then there exist integers r and s such that $rm + sn = d$ (prove this as an exercise). By Lemma 2 part (i), we can write $2F_d = F_{rm} L_{sn} + F_{sn} L_{rm}$. Let $(F_m, F_n) = h$. Then $h | F_m$ and $h | F_n$. By Lemma 3 part (iii), $h | F_{rm}$ and $h | F_{sn}$. So $h | (F_{rm} L_{sn} + F_{sn} L_{rm}) = 2F_d$. If h is odd, $h | F_d$. If h is even, F_m, F_n, F_{sn} , and F_{rm} are all even, based on the statements above. So now we can write $F_d = F_{rm} (\frac{1}{2} L_{sn}) + F_{sn} (\frac{1}{2} L_{rm})$. Again, $h | F_d$, and therefore $h | F_d$ in all cases. But from part (iii) we know that $F_d | F_m$ and $F_d | F_n$ since m and n are multiples of d . So $F_d | (F_m, F_n) = h$. Thus, $h | F_d$ and $F_d | h$, so $h = F_d$. Q.E.D.

In order to finish proving our conjecture, we need to use two more topics in number theory: Fermat's Little Theorem and the Legendre symbol.

Lemma 4: Fermat's Little Theorem. For integer a and prime p such that $(a, p) = 1$, $a^{p-1} \equiv 1 \pmod{p}$.

Proof: The case $a = 1$ is trivial, so consider $a \neq 1$. Then consider the sequence $a, 2a, \dots, (p-1)a$. Each term must be distinct from the others in mod p (if xa and ya are congruent, then $(x-y)p$, which is impossible unless $(x-y) = 1$, which in turn implies that $a | p$, which is also impossible). Since there are $p-1$ distinct terms in the sequence and there are $p-1$ least residues mod p , each term must correspond to one and only one residue mod p . Then consider the product of all the terms of the sequence and the product of all the residues in mod p . This will yield $(p-1)! \cdot a^{p-1} \pmod{p} \equiv (p-1)! \pmod{p}$, and thus $a^{p-1} \equiv 1 \pmod{p}$. Q.E.D.

Before defining the Legendre symbol, we shall first discuss quadratic residues. k is a quadratic residue of p if and only if there exists some integer x such that $x^2 \equiv k \pmod{p}$. There are many theorems regarding whether a number is a quadratic residue or not in a given prime modulus. For example, 2 is a QR (quadratic residue) of all primes p where $p \equiv 1 \pmod{8}$, 3 is a QR for all $p \equiv 1 \pmod{12}$, 5 is a QR for all $p \equiv 1 \pmod{5}$ and so on. We will use the theorem regarding when 5 is a QR in mod p later to prove our conjecture.

The Legendre's symbol $\left(\frac{a}{p}\right)$ is defined as 1 if a is a quadratic residue mod p , and -1 if a is not a quadratic residue mod p . The theorem that we will use to prove our conjecture is Euler's Lemma (prove as an exercise).

Euler's Lemma. If p is prime greater than 2 and a is a residue in mod p , then $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$.

Now we can finally prove our initial conjecture.

Theorem. For prime p , $F_{p-1} \equiv 0 \pmod{p}$ if $p \equiv 1 \pmod{5}$, and $F_{p+1} \equiv 0 \pmod{p}$ if $p \equiv 2 \pmod{5}$.

Proof. Since $F_3 = 2$ and $F_5 = 5$, we can assume that $p \neq 2$ and $p \neq 5$. Then consider $2^{n-1} F_n$. By Theorem 1,

$$2^{n-1} F_n = \left(\frac{2^{n-1}}{\sqrt{5}}\right) \left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n \right] = \left(\frac{1}{2\sqrt{5}}\right) \left[(1+\sqrt{5})^n - (1-\sqrt{5})^n \right].$$

Using the binomial theorem,

$$2^{n-1}F_n = \left(\frac{1}{2\sqrt{5}}\right) \left[\left(\binom{n}{0}1 + \binom{n}{1}\sqrt{5} + \binom{n}{2}(\sqrt{5})^2 + \dots + \binom{n}{n}(\sqrt{5})^n \right) - \left(\binom{n}{0}1 - \binom{n}{1}\sqrt{5} + \binom{n}{2}(\sqrt{5})^2 - \dots + (-1)^n \binom{n}{n}(\sqrt{5})^n \right) \right]$$

$$\text{If } n \text{ is even, } 2^{n-1}F_n = \left(\frac{1}{2\sqrt{5}}\right) \left[2\binom{n}{1}\sqrt{5} + 2\binom{n}{3}(\sqrt{5})^3 + \dots + 2\binom{n}{n-1}(\sqrt{5})^{n-1} \right] = n + \binom{n}{3}5 + \binom{n}{5}5^2 + \dots + n\binom{n}{\frac{n-1}{2}}5^{\frac{n-1}{2}}.$$

$$\text{If } n \text{ is odd, } 2^{n-1}F_n = \left(\frac{1}{2\sqrt{5}}\right) \left[2\binom{n}{1}\sqrt{5} + 2\binom{n}{3}(\sqrt{5})^3 + \dots + 2\binom{n}{n-1}(\sqrt{5})^{n-1} \right] = n + \binom{n}{3}5 + \binom{n}{5}5^2 + \dots + 5^{\frac{n-1}{2}}.$$

Consider this expression when n is a prime number (let $n = p$ for convenience). $2^{p-1} \equiv 1 \pmod{p}$ by Fermat's

Little Theorem. Also, for $p > 2$, p is odd. Then, $2^{p-1}F_p \equiv F_p \equiv p + \binom{p}{3}5 + \binom{p}{5}5^2 + \dots + \binom{p}{\frac{p-1}{2}}5^{\frac{p-1}{2}} \pmod{p}$. Since

the binomial coefficients (except for the last term) are divisible by p , $F_p \equiv 5^{\frac{p-1}{2}} \equiv \left(\frac{5}{p}\right) \pmod{p}$ by

Euler's Lemma and the definition of Legendre's symbol.

Using Theorem 2 part (iii), $(F_p)^2 - F_{p-1}F_{p+1} = (-1)^{p-1}$. Since p is odd, $p-1$ is even and $(-1)^{p-1}$ is equal to 1. Using $F_p \equiv 1 \pmod{p}$, $F_{p-1}F_{p+1} \equiv 0 \pmod{p}$. Also, $(p-1, p+1) = 2$, so $(F_{p-1}, F_{p+1}) = F_2 = 1$ by Theorem 3 part (iv). Therefore only one of the two terms F_{p-1}, F_{p+1} is divisible by p . Now we take $n = p+1$. Since $p+1$ is

even, then $2^p F_{p+1} = (p+1) + \binom{p+1}{3}5 + \binom{p+1}{5}5^2 + \dots + (p+1)5^{\frac{p-1}{2}}$. If we look at $2^p F_{p+1} \pmod{p}$, again all the terms of the binomial expansion drop out except the first and the last term. So $2^p F_{p+1} \equiv 2F_{p+1} \equiv (p+1) + (p+1)5^{\frac{p-1}{2}} \equiv (p+1)(1 + \left(\frac{5}{p}\right)) \equiv 1 + \left(\frac{5}{p}\right) \pmod{p}$. Therefore $F_{p+1} \equiv 0 \pmod{p}$ if and only if $\left(\frac{5}{p}\right) = -1$. Accord-

ing to a theorem regarding quadratic residues (prove as an exercise) $\left(\frac{5}{p}\right) = -1$ if and only if $p \equiv 2 \pmod{5}$.

If $n = p-1$, we can similarly prove that $F_{p-1} \equiv 0 \pmod{p}$ if and only if $\left(\frac{5}{p}\right) = 1$, or $p \equiv 1 \pmod{5}$. Q.E.D.

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Determining Probabilities Using Markov Chains

John Choi

Andrey Andreyevich Markov (1856-1922) was a Russian mathematician whose early work centered around number theory and analysis. His later works dealt with the study of mutually dependent variables, and in the course of his studies Markov proceeded to develop the important concept of chained events, which he represented using Markov chains.

A Markov chain is a sequence of events that must have the following:

1. A set of states. The process goes from one state to another, but we only look at a single state at a time.
2. A set of transition probabilities p_{ij} . Each p_{ij} gives the probability of moving from a state i from a state j .
3. An initial probability distribution $\{p_o(i)\}$ where $p_o(i)$ gives the probability of state i at the time the process begins.

When we work with Markov chains, we often use transition matrices. A transition matrix by definition, is a matrix whose entries (the transition probabilities) are the probability of moving from one state j to a state i . Such a matrix must have the elements of its rows adding up to one, and it must be a square matrix (having an equal number of rows as it does columns).

Let's take a quick look at how a transition matrix is used in the following problem.

A yellow taxi in Manhattan just dropped off another satisfied New Yorker at the World Trade Center in downtown Manhattan. Our driver, John, is on his lunch break and decides to take a scenic tour of the island of Manhattan. However, he is divided in opinion on where to go. He could drive around downtown NYC, Central Park (in central NYC), or spend his time uptown. John has an equal probability of staying in one place, or moving up or down one place. In other words, John cannot skip passing through central Manhattan. What is the probability, after two trips, that John will be back in downtown Manhattan?

Since we know that the driver begins downtown Manhattan, he has two choices: he could drive around there for a while, or he can go to Central Park. To put it in terms of fractions, John has a $1/2$ probability of staying where he is and a $1/2$ probability of moving up to Central Park. If he decides to move to Central Park, then he has $1/3$ chance to stay where he is, $1/3$ chance to go uptown, and $1/3$ chance to go back downtown. The probability of his staying downtown for the two trips is $1/4$, and the probability of his driving to midtown first and then back downtown is $1/6$, so his total probability after two trips of ending up downtown is $5/12$.

An alternative method of calculating these probabilities is to use Markov chains. M_1 , at left, is the transition matrix, which shows the probabilities of his changing position (following the second condition of the Markov chains). For example, to obtain the probability that John drives up to Central Park and back, we should multiply P_{dc} by P_{cd} , which is equal to $1/6$. The probability that he merely stays downtown, $P_{dd} \cdot P_{dd}$, is $1/4$, so the sum of the two probabilities is $5/12$. This is really just the same process as the above.

This is the transition matrix M_1^2 , which displays the probabilities of the taxi being in a location after two states, or trips. We are specifically concerned with P_{dd}^2 , which shows the total probability his being downtown after two trips, $5/12$. Reading the probability from the squared transition matrix is more convenient than having to calculate all the intermediate probabilities.

Another example of a Markov chain is the "gambler's ruin".

Two gamblers, Monte and Karla, play a series of games of chance with \$1 at stake each time. In each game the probability that Monte will win is $1/4$ and the probability that Monte will lose is $3/4$, and no draws are allowed. The game ends when one of the gamblers loses all his or her money. If Monte begins the game with \$3

and Karla begins with \$1, find the probability that he will lose all his money after 4 games.

The possible states of this question are 0 (when Monte has \$0), 1, 2, 3, and 4 (when Monte has won all of Karla's dollars and now has \$4). The states of the Markov chain are in terms of Monte's money. The probability of p_{00} is 1, because in state 0, Monte has no money, can no longer win any money, and will continue to have \$0. So, we can assume that when Monte has reached state 0, he will be ruined and the game ends. The same logic is used for state 4, where Monte has all of Karla's money and the game ends. So, we have our transition matrix T (at right):

$$T = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & \frac{3}{4} & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{3}{4} & 0 \\ 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & 1 \end{bmatrix} \end{matrix}$$

We know more than just the transition probabilities. We are also given the initial position of the player, which state 3 (he begins with \$3). We can now specify the initial probability vector. In general, the initial

probability vector of a Markov chain with m possible states is the $m \times 1$ matrix: $Q = \begin{bmatrix} q_1 \\ \vdots \\ q_m \end{bmatrix}$, where q_i is the probability that the system is in state i initially. Going back to the problem, the initial probability vector is $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$.

This is because we know the system is in state 3. Therefore, the probability of being in initial state 3 is 1, and the probability of being in any other state initially is 0, which is not the least bit surprising.

We can extend the idea of an initial probability vector to later states if we let $Q^{(k)}$ be the $m \times 1$ matrix such that $q_i^{(k)}$ is the probability that the system is in state i after k transitions. Then $Q^{(k)}$ is called the k th-step probability vector. In order to solve my problem concerning our gamblers, we must find the k th step probability vector of Monte in four transitions. How do we find this seemingly elusive vector?

There is a theorem that states that if P is the transition matrix of a Markov chain with initial probability vector Q and k th step probability vector $Q^{(k)}$, then: $Q^{(k)} = P^k Q$.

In this case, we multiply the initial probability vector by the fourth power of the initial transition matrix to obtain the k th step probability vector.

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & .890625 & .773438 & .421875 & 0 \\ 0 & .070313 & 0 & .210938 & 0 \\ 0 & 0 & .140625 & 0 & 0 \\ 0 & .023448 & 0 & .070313 & 0 \\ 0 & .015625 & .085938 & .296875 & 1 \end{bmatrix} = \begin{bmatrix} .421875 \\ .210938 \\ 0 \\ .070313 \\ .296875 \end{bmatrix}$$

Thus, the probability of Monte's going broke after four games is .421875, or 27/64.

At this point, we have dealt with the finite aspects of Markov chains (the taxi problem and the gambler's ruin). Let's take a look into infinite Markov processes, the long-term effects of Markov chains.

Theorem 1. *If a Markov chain has a regular transition matrix P , then there exists a vector s with the property that for any initial probability vector q , the sequence of vectors $q, Pq, P^2q, \dots$ approaches s , and s is called the steady-state vector. Moreover, s satisfies the equation $Ps = s$ and is the only probability vector that does so (the proofs for both of these statements is beyond the scope of this article).*

Observe how we find the steady state vector in the problem given below.

There are two famous restaurants in Little Italy, Luigi's World-Famous Pizza and Ray's Best Pizza. Mario, who just moved into the neighborhood, likes to have a pizza party once a month. Because Luigi was the name of Mario's favorite uncle, Mario has a 75% chance of going to Luigi's first. If Mario goes to Luigi first, there is a 60% chance of returning next time. If he goes to Ray's, there is a 80% he will not return next time. What is the probability that Mario will go back to Luigi's World-Famous Pizza in the long run?

Here our initial probability vector is $\begin{bmatrix} 0.75 \\ 0.25 \end{bmatrix}$, but we can disregard it because by Theorem 1, the steady-state vector is not affected by the initial probability vector. The initial transition matrix is $P = \begin{bmatrix} 0.6 & 0.8 \\ 0.4 & 0.2 \end{bmatrix}$. Taking the transition matrix P , we know that there exists a steady-state vector $\begin{bmatrix} p \\ q \end{bmatrix}$ that expresses the long-term behavior of our friend Mario. This vector must also satisfy $p+q=1$ (since the steady-state vector's entries represent probabilities, all of its elements must add to one). and it is the unique solution to the equation.

$$\begin{bmatrix} 0.6 & 0.8 \\ 0.4 & 0.2 \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}$$

So, by multiplying the matrices, we obtain the system of equations:

$$0.6p + 0.8q = p$$

$$0.4p + 0.2q = q$$

Solving for p ,

$$0.8q = 0.4p$$

$$0.4p = 0.8q \quad 2q = p !$$

Thus, the steady-state vector is $\begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}$, and the probability that Mario will be at Luigi's in the long run is exactly $2/3$.

Markov chains can be applied to a wide variety of real-world problems. In genetics, Markov chains can be used to determine the probability that a genetic disease will be passed from a person to their offspring. Different states can be assigned to each individual generation. A biologist can also use Markov chains to determine population migrations of animals. This can be accomplished, given the probability that each population will stay in or migrate from a particular region. In chemistry, if a given compound is energized by an electric charge, it can either mutate into a specific number of compounds, or it can stay the same. A chemist can use Markov chains to find the different probabilities of each case occurring.

In addition to all these applications, Markov chains can be used to solve the day-by-day questions in life, as shown in the taxi problem and Mario's problem.

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RSA Cryptography

Soojin Yim

Consider the following: I have two large distinct prime numbers p and q which are known only to myself, and a secret number N relatively prime to the product $(p-1)(q-1)$. The numbers that I make public are the product pq and a positive number M satisfying $MN \equiv 1 \pmod{(p-1)(q-1)}$. I make them public by writing them in a public directory, and anyone who wants to send me a message, represented by a number x , should transmit to me the least nonnegative residue of $y = x^M \pmod{pq}$.[†] When I receive a number y , which I know is the encrypted form of part of someone's message, I decipher it by calculating the least nonnegative residue of $y^N \pmod{pq}$, where N is my secret exponent. This will yield the original number x that my correspondent wished to convey to me.

For example, my secret numbers are $p = 11$, $q = 13$, and $N = 7$ (these are small numbers, for pedagogical purposes). I privately calculate, using the Euclidean algorithm, that the smallest positive number M that satisfies $7M \equiv 1 \pmod{120}$ is $M = 103$.[‡] Then, I publicly announce that anyone who wants to send me a message x , should transmit to me the least nonnegative residue of $x^{103} \pmod{143}$. If the message x is represented by the number 19, then the person who wishes to send me a message calculates $19^{103} \equiv 72 \pmod{143}$, and sends me the 72, the encrypted message y . When I receive that message, I calculate $72^7 \equiv 19 \pmod{143}$, and so I know the original message was 19.

By now, you may be wondering: why exactly does this method work? What is happening is that a number is enciphered by raising it to the M th power mod pq , and it is deciphered by raising the resulting message by the N th power mod pq to retrieve the original message. If x is the number representing the message, then $y = x^M \pmod{pq}$ is the encrypted message, and $y^N = (x^M)^N \pmod{pq}$ is the deciphering of the encrypted message — meaning, this equals x .

Theorem. If p and q are distinct prime numbers and $MN \equiv 1 \pmod{(p-1)(q-1)}$, then $(x^M)^N \equiv x \pmod{pq}$.

Proof. Choose a positive integer k so that $MN = k(p-1)(q-1) + 1$, which can be done since $MN \equiv 1 \pmod{(p-1)(q-1)}$. Thus, $(x^M)^N = x^{MN} = x^{k(p-1)(q-1)+1} = x^{k(p-1)(q-1)} \cdot x$, and so we need to show that $x^{k(p-1)(q-1)} \cdot x \equiv x \pmod{pq}$. There are three cases:

Case 1. Suppose x is divisible by both p and q . Then both sides of the equation are congruent to 0 mod pq , and the claim is true.

Case 2. Suppose x is divisible by q but not by p . (A similar argument works if x is divisible by p but not q). Fermat's Little Theorem states that for integer a and prime p , $a^{p-1} \equiv 1 \pmod{p}$. If $a = x^{k(q-1)}$ then $(x^{k(q-1)})^{p-1} \equiv 1 \pmod{p}$. Because a congruence relation can be preserved by multiplying both sides (and the modulus) by the same number, multiplying by q the expression becomes $x^{k(p-1)(q-1)} \cdot q \equiv q \pmod{pq}$. Since x is assumed to be a multiple of q , we can write $x = qx'$, for some integer x' . Multiplying both sides by x' yields $x^{k(p-1)(q-1)} \cdot qx' \equiv qx' \pmod{pq}$, which simplifies to $x^{k(p-1)(q-1)} \cdot x \equiv x \pmod{pq}$.

Case 3. Suppose x is not divisible by p or q . By Fermat's Little Theorem, if $a = x^{k(q-1)}$ then $(x^{k(q-1)})^{p-1} \equiv 1 \pmod{p}$, and similarly, $(x^{k(p-1)})^{q-1} \equiv 1 \pmod{q}$. From these two equations, we know that $x^{k(p-1)(q-1)} - 1$ is a multiple of both p and q . Since p and q are distinct primes, this implies that $x^{k(p-1)(q-1)} - 1$ is a

[†] The least nonnegative residue of number $a \pmod{m}$ is the remainder r , $0 \leq r \leq m-1$, when a is divided by m .

[‡] The Euclidean algorithm works as follows: $120/7 = 17 \text{ R } 1$. Thus, $1 = 120 - 17 \times 7$ shows that $(-17) \times 7 \equiv 1 \pmod{120}$, so M could be -17 . However, because M must be positive, we convert -17 to 103 in mod 120 arithmetic.

multiple of pq ($x^{k(p-1)(q-1)} - 1 \equiv 0 \pmod{pq}$). Hence, $x^{k(p-1)(q-1)} \cdot x \equiv x \pmod{pq}$.

This method is known as RSA cryptography, named after its inventors (namely, Rivest, Shamir, and Adleman). Not only have we established that the RSA method works, we have proven *why* this method works. However, it should be reiterated that this method is only secure for primes p and q that are so large that their product pq is essentially impossible to factor, or at least impossible in an acceptable time period. With the above example, someone could easily have figured out from the public key 143, the private keys 11 and 13, and thus destroy the security of the system. This would be done once the secret exponent N is found, allowing the intruder to solve the equation $130N \equiv 1 \pmod{120}$. Using the Euclidean algorithm, $N = 7$,[†] and now all the private numbers are known. Armed with this knowledge, anyone who intercepts the messages can decipher them. Currently, integers p and q should be about 100 digits each to make factoring pq impractical.

From Fermat's simple theorem, the difficulty of prime factorization, the cryptography method that defies even today's fast computers emerged. Thanks to a 17th century mathematician, we in the 1990s are able to send secure messages without fear of interception and decryption.

[†] Similarly, here we begin with $17 = 120 - 1 \cdot 103$, and $1 = 103 - 6 \cdot 17$. Combining them yields $1 = 103 - 6(120 - 103) = 7 \cdot 103 - 6 \cdot 120$, and so $N = 7$.

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Infinity and Cardinality

Amy Yang

Infinity is a term referring to a quantity so large that it cannot be measured in any simple sense. Despite its illusory nature, infinity plays an important role in the theory of sets. In 1638, the famous Italian scientist Galileo realized that the set of natural numbers $\{1, 2, 3, \dots\}$ is in a direct one-to-one correspondence with the set of their squares, $\{1, 4, 9, \dots\}$. He reasoned that since each element of the second set can be linked to one and only one element of the first, both sets must have the same number of elements. However, this was problematic: intuitively, the set of squares is a subset of the natural numbers, so surely the first list contained more numbers than the second one, not an equal number.

In Germany, at the turn of the century, the Russian-born mathematician Georg Cantor applied the tools of mathematical rigor and logical deduction to the concept of infinity. Cantor, the founder of the branch of mathematics called set theory, defined a set as a collection of well-distinguished and well-defined objects. He studied infinite sets (sets which have an infinite number of members) and came up with surprising results.

Cantor's arithmetic was very simple: two infinite sets are equal in number if they can be put in a one-to-one correspondence with each other. For example:

Set of Natural Numbers	$\leftrightarrow$	Set of Squares
1	$\leftrightarrow$	1
2	$\leftrightarrow$	4
3	$\leftrightarrow$	9
...	$\leftrightarrow$	...

Sets that can be matched to each other as in this example are said to have the same *cardinality*. Whenever a set has a one-to-one correspondence with the set of natural numbers, the set is said to be *denumerable* or *countably infinite* and their cardinality is denoted by the Hebrew letter aleph-null, $\aleph_0$. However, the set of natural numbers is infinite only to a certain extent, thus having a *transfinite* cardinal number.

There is an interesting illustration of the consequences of Cantor's proof of the fact that parts of infinity may be in fact be equal to the infinite whole. It is a little story devised by Hilbert that has become known as "Hilbert's Hotel":

Let us imagine a hotel with a finite number of rooms, and assume that all the rooms are occupied. Obviously, a new guest will not be able to fit. Now let us imagine a hotel with a countably infinite number of rooms, but that is also completely occupied. This time, when a new guest arrives, the proprietor can move the person in room 1 into room 2, the person previously occupying room 2 into room 3, and so on ... and the new customer receives room 1, which becomes free as a result of these transpositions. Let us imagine now that a countably infinite number of guests comes to this infinite hotel and asks for rooms. The proprietor then moves the occupant of room 1 into room 2, the occupant of room 2 into room 4, the occupant of room 3 into room 6, and so on. Now all the odd-numbered rooms are free and the infinity of new guests can be accommodated in them. Therein lies the power of infinite sets.



Solution to puzzle on page 40

M.C. Escher

Edmund Chou

Maurits Cornelis Escher was born in 1898 in Leeuwarden, Holland. Although his father was a civil engineer, and three of his four brothers pursued science or engineering, Escher was a poor mathematics student. With his high school art teacher's encouragement, he soon became interested in graphic arts.

Escher has always felt as if he had an inability to understand mathematics, referring to himself as "absolutely innocent of training or knowledge in the exact sciences." However, he had been fascinated by order and symmetry since his childhood, leading him to study the tiling patterns in the Alhambra castle in Granada, to examine geometric diagrams in mathematical papers, and to create his own unique ideas for tiling a plane (in which several shapes, or "tiles," can be repeated in an endless pattern, covering an infinite plane surface).

Escher entered the School for Architecture and Decorative Arts in Haarlem in 1919 to study architecture. After seeing Escher's works, an impressed artist, Samuel Jessurun de Mesquita, invited him to concentrate in graphic arts. De Mesquita had a great influence on Escher as a teacher, friend, and fellow artist.

Escher settled in Rome after his studies in Haarlem, where he toured and sketched extensively, particularly in southern Italy. His eyes were able to discern striking visual effects from the ordinary. His works included architectural details of monumental buildings from unusual vantage points, light and shadows cast by warrens of staircases in tiny villages, clusters of homes clinging to mountain slopes that plunged into distant valleys, and tiny details of nature viewed as if through a magnifying glass. His sketches were later made into woodcuts or lithographs in his studio.

When the political climate became unbearable in 1935, Escher left Italy with his wife and young sons. They lived in Switzerland for two years, then three years in Uccle, near Brussels, and then settled permanently in Baarn, Holland. This period brought a change in Escher's work: his inspiration came no longer from his eyes, but from within his mind. His goal became to give visual expression to concepts and to portray the complexities of human observation and understanding. When trying to reach this goal, he often found himself in a world governed by mathematics.

Escher was extremely interested with the "regular division of the plane." He created over 150 infinite tilings, in which figures crawled, swam, soared, and filled the plane with their clones. These figures also exhibited symmetries of many different kinds. Escher felt that these planar divisions also illustrated the concept of infinity; some of his drawings or "tiles" can be continued indefinitely. Thus, Escher challenged himself to "contain" infinity within a single page.

Escher wrote, "Anyone who plunges into infinity, in both time and space, farther and farther without stopping, needs fixed points, mileposts as he flashes by, for otherwise his movement is indistinguishable from standing still. He must mark off his universe into parts of a certain length, into compartments which repeat one another in endless succession."



From M.C. Escher and J.C. Locher, *The World of M.C. Escher* (New York: Harry Adams, 1972).

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Math Survey AP Calculus Review

Originally by David Z. Weinstein, Updated by Math Survey Staff

All material is covered by BC calculus; asterisks (*) indicate material that does not apply to AB calculus.

Limits

1.* Defn: Let $f(x)$ be defined in some open interval containing a (except possibly at a itself), then

$\lim_{x \rightarrow a} f(x) = L$, if for any $\epsilon > 0$ (however small) there exist a $\delta > 0$ such that $|f(x) - L| < \epsilon$ ($L - \epsilon < f(x) < L + \epsilon$) whenever $0 < |x - a| < \delta$ ($a - \delta < x < a + \delta$).

a.* Defn: The statement $\lim_{x \rightarrow \infty} f(x) = L$ means that for every $\epsilon > 0$, there exists an $M > 0$ such that $|f(x) - L| < \epsilon$ whenever $x > M$.

b.* Defn: The statement $\lim_{x \rightarrow -\infty} f(x) = L$ means that for every $\epsilon > 0$, there exists an $N < 0$ such that $|f(x) - L| < \epsilon$ whenever $x < N$.

c.* Defn: The statement $\lim_{x \rightarrow c} f(x) = \infty$ means that for each $M > 0$, there exists a $\delta > 0$ such that $f(x) > M$ whenever $0 < |x - c| < \delta$.

d.* Defn: The statement $\lim_{x \rightarrow c} f(x) = -\infty$ means that for each $N < 0$, there exists a $\delta > 0$ such that $f(x) < N$ whenever $0 < |x - c| < \delta$.

2. Limit May Fail to Exist if

a. $\lim_{x \rightarrow a^+} f(x) \neq \lim_{x \rightarrow a^-} f(x)$ (e.g. $\lim_{x \rightarrow 0} \frac{|x|}{x}$).

b. values of $f(x) \rightarrow \infty$ (e.g. $\lim_{x \rightarrow 0} \frac{1}{x}$).

c. values of $f(x)$ may oscillate indefinitely, approaching no limit (e.g. $\lim_{x \rightarrow 0} \sin \frac{1}{x}$).

3. Basic Limit Theorems:

a. $\lim [f(x) \pm g(x)] = \lim f(x) \pm \lim g(x)$.

b. $\lim [f(x) \cdot g(x)] = \lim f(x) \cdot \lim g(x)$.

c. $\lim [f(x)/g(x)] = \lim f(x)/\lim g(x)$ ($\lim g(x) \neq 0$).

d. $\lim [\sqrt[n]{f(x)}] = \sqrt[n]{\lim f(x)}$.

Continuity

1. Defn: f is continuous at a provided $f(a)$ is defined, $\lim_{x \rightarrow a} f(x)$ exists, and $\lim_{x \rightarrow a} f(x) = f(a)$ (equivalently: $f(x)$ is continuous at $x = a$, if for any $\epsilon > 0$, there exists a $\delta > 0$, such that $|f(x) - f(a)| < \epsilon$ whenever $|x - a| < \delta$).

2. Important Limit Theorems

a. If f and g are continuous functions, then fg , cf , $f \pm g$, f/g ($g \neq 0$) are also continuous.

b. All polynomials are continuous.

- c. A function continuous on a closed interval $[a, b]$ has the following properties.
- $f(x)$ has a minimum and maximum value on the interval.
 - $f(x)$ cannot pass from one value to another in the interval without taking every intermediate value (i.e. It's an unbroken line).

Derivatives

1. Defn: let $x = a$ be on the curve $y = f(x)$. At each such point where $\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ exists, the function is said to be differentiable (have a derivative). The limit is called the derivative of y with respect to x and denoted by $\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$. Additionally, $f'(a) =$

$$\lim_{\Delta x \rightarrow 0} \frac{f(a + \Delta x) - f(a)}{\Delta x} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \text{ (derivative at } a\text{). If } y = f(x) \text{ then}$$

- $dy/dx = f'(x) = D_x y = y'$ (first derivative).
 - $d^2y/dx^2 = f''(x) = D_x^2 y = y''$ (second derivative).
 - $d^3y/dx^3 = f'''(x) = D_x^3 y = y'''$ (third derivative).
2. Formulas for Derivatives
 c, n are constants; u, v are differentiable functions of x .

- | | | | |
|--|--|---|--|
| 1. | 2. | 3. | 4. |
| $\frac{d(c)}{dx} = 0$ | $\frac{d(cu)}{dx} = c \frac{du}{dx}$ | $\frac{d(u^n)}{dx} = nu^{n-1} \frac{du}{dx}$ | $\frac{d(u+v)}{dx} = \frac{du}{dx} + \frac{dv}{dx}$ |
| 5. | 6. | 7. | 8. |
| $\frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$ | $\frac{d\left(\frac{u}{v}\right)}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ | $\frac{d(\sin u)}{dx} = \cos u \frac{du}{dx}$ | $\frac{d(\cos u)}{dx} = -\sin u \frac{du}{dx}$ |
| 9. | 10. | 11. | 12. |
| $\frac{d(\tan u)}{dx} = \sec^2 u \frac{du}{dx}$ | $\frac{d(\cot u)}{dx} = -\csc^2 u \frac{du}{dx}$ | $\frac{d(\sec u)}{dx} = \sec u \tan u \frac{du}{dx}$ | $\frac{d(\csc u)}{dx} = -\csc u \cot u \frac{du}{dx}$ |
| 13. | 14. | 15. | 16. |
| $\frac{d(\ln u)}{dx} = \frac{1}{u} \frac{du}{dx}$ | $\frac{d(e^u)}{dx} = e^u \frac{du}{dx}$ | $\frac{d(\log_e u)}{dx} = \log_e e \cdot \frac{1}{u} \frac{du}{dx}$ | $\frac{d(a^u)}{dx} = a^u \ln a \frac{du}{dx}$ |
| 17. | 18. | 19. | 20. |
| $\frac{d(\sin^{-1} u)}{dx} = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$ | $\frac{d(\cos^{-1} u)}{dx} = -\frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$ | $\frac{d(\tan^{-1} u)}{dx} = \frac{1}{1+u^2} \frac{du}{dx}$ | $\frac{d(\cot^{-1} u)}{dx} = -\frac{1}{1+u^2} \frac{du}{dx}$ |
| 21. | 22. | | |
| $\frac{d(\sec^{-1} u)}{dx} = \frac{1}{\sqrt{u^2-1}} \frac{du}{dx}$ | $\frac{d(\csc^{-1} u)}{dx} = -\frac{1}{\sqrt{u^2-1}} \frac{du}{dx}$ | | |

23. Chain Rule (for composite functions)

If $y = f(u)$ and $u = g(x)$ ($y = f(g(x))$), then $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = f'(u)g'(x) = f'(g(x))g'(x)$.

24. If $y = f(x)$ is single-valued and has nonzero derivatives on a given interval, then the inverse function exists and has a derivative. Suppose $y = f(x)$ has an inverse function (i.e. $x = f^{-1}(y)$ exists) that is

differentiable, then $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$.

25. *Parametric Equations

If $y = f(t)$ and $x = g(t)$, then

a. * $y' = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.

b. * $y'' = \frac{d^2y}{dx^2} = \frac{dy'}{dx} = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}}$.

c. * $y''' = \frac{d^3y}{dx^3} = \frac{dy''}{dx} = \frac{\frac{dy''}{dt}}{\frac{dx}{dt}}$.

26. Logarithmic Differentiation

If $y = x^{g(x)}$ then $\ln y = \ln x^{g(x)} = g(x) \ln x$. Differentiating yields $\frac{1}{y} \frac{dy}{dx} = g'(x) \ln x + g(x) \frac{1}{x}$ or

$$\frac{dy}{dx} = y \left(g'(x) \ln x + g(x) \frac{1}{x} \right).$$

Curve Sketching

1. Without Derivatives

- Intercepts: To get the x intercepts, set $y = 0$ and solve for x ; to get the y intercepts, set $x = 0$ and solve for y .
- Symmetry: If a curve has form $y = F(x)$ or $f(x, y) = 0$, then it is
 - Symmetric to x -axis if $f(x, y) = f(x, -y)$ (or $F(x) = -F(x)$).
 - Symmetric to y -axis if $f(x, y) = f(-x, y)$ (or $F(x) = F(-x)$).
 - Symmetric to origin if $f(x, y) = f(-x, -y)$ (or $F(x) = -F(x)$).
- Extent (domain and range): Find x and y values on which the function is defined.
- Asymptotes:
 - Horizontal Asymptote: Let $x \rightarrow \pm\infty$ and find the limiting value(s) of y .
 - Vertical Asymptote: Let $y \rightarrow \pm\infty$ and find the limiting value(s) of x .
 - Oblique Asymptote $y = mx + b$ ($m \neq 0$) is an asymptote of $y = f(x)$ if

$$\lim_{x \rightarrow \infty} [f(x) - (mx + b)] = 0.$$

- We can also solve for one variable in terms of the other; set denominator equal to 0, etc. For

example, if $y = f(x) = g(x) + \frac{c}{h(x)}$, then $y = g(x)$ is a curved (e.g. parabolic, oblique, or horizontal) asymptote, and $h(x) = 0$ represents any vertical asymptotes.

2. With Derivatives

a. Defn:

- i. A function $f(x)$ is increasing (decreasing) on an interval if whenever x_1 and x_2 belong to the interval and $x_2 > x_1$, then either $f(x_2) > f(x_1)$ (rising curve) or $f(x_2) < f(x_1)$ (falling curve).
- ii. $f(x)$ is a smooth function if both $f(x)$ and $f'(x)$ are continuous.
- iii. Relative Extremum: maximum or minimum value.
- iv. Critical values of $f(x)$ are values where $f'(x) = 0$ or is undefined.
- v. A smooth curve is concave upward at a point if successive tangents to the curve from that point turn counterclockwise (slope increases - curve lies above tangent). A smooth curve is concave downward at a point if successive tangents to the curve from that point turn clockwise (slope decreases - curve lies below tangent).
- vi. Point of inflection: a point where the curve changes its concavity.



b. Important Theorems:

- i. Function $f(x)$ is increasing at $x = a$ if $f'(a) > 0$, decreasing at $x = a$ if $f'(a) < 0$.
- ii. If $f(x)$, a continuous, differentiable function, has a relative extrema at $x = a$, then $f'(a) = 0$ (but not conversely).

iii. Tests for Maxima and Minima

- (1) Max: $f'(a) = 0$ and $f'(x)$ changes from + to - at $x = a$.
Min: $f'(a) = 0$ and $f'(x)$ changes from - to + at $x = a$.
 - (2) Max: $f'(a) = 0$ and $f''(a) < 0$.
Min: $f'(a) = 0$ and $f''(a) > 0$.
 - iv. Curve is concave up at $x = a$ if $f''(a) > 0$; concave down if $f''(a) < 0$. (If $f''(a) = 0$, it's concave up or down depending on its concavity in the neighborhood of $x = a$.)
 - v. If $f(x)$ is a function with continuous second derivative, $x = a$ is a point of inflection of $f(x)$, if $f''(a) = 0$ and $f''(a) \neq 0$ or $f''(x)$ changes sign through a .
- c. Absolute extrema can be attained
- i. Where $f'(x) = 0$.
 - ii. At the endpoints of interval.
 - iii. Where the derivative does not exist.



Applications of the Derivative

1. If a function is differentiable at a point, then it must be continuous at the point, but not conversely, i.e.: differentiability implies continuity.
2. Rolle's theorem: If $f(x)$ is continuous on $[a, b]$, $f'(x)$ exists on (a, b) , and $f(a) = f(b) = 0$, then there exists a point c ($a < c < b$) such that $f'(c) = 0$.
3. Mean value theorem for derivatives: If $f(x)$ is continuous on $[a, b]$ and $f'(x)$ exists on (a, b) then there

exists a point c such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

4. L'Hôpital's Rule

- a. If $f(x)$ and $g(x)$ are in an interval including $x = a$ and satisfy the hypotheses of the general mean

value theorem and $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$.

- b. Evaluation of other indeterminate forms:

i. $0 \cdot \infty$. Change to $(1/\infty)(\infty)$, which equals ∞/∞ .

ii. $\infty - \infty$. Transform by algebraic or trigonometric manipulation to acceptable form.

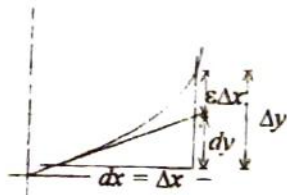
iii. 1^∞ , 0^0 , and ∞^0 . Take the log of the function and change to one of the above forms.

5. Differentials. Defn: When $y = f(x)$, the differential of x (independent value) $dx = \Delta x$ and the differential of y (dependent value) $dy = f'(x) dx$.

Motivation for definition:

- a. One may now think of $f'(x)$ as a quotient.

- b. For small Δx , $dy = \Delta y$.



Note: $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$, therefore $\frac{\Delta y}{\Delta x} = f'(x) + \varepsilon$ ($\varepsilon \rightarrow 0$ when $\Delta x \rightarrow 0$)

implies $\Delta y = f'(x)\Delta x + \varepsilon\Delta x$. Main part of Δy is $f'(x)\Delta x$ since $\varepsilon\Delta x$ is relatively small. So we define $dy = f'(x)\Delta x$, therefore $\Delta y = dy + \varepsilon\Delta x \approx dy$.

6. Related rates problems: Find the relationship between variables, take the derivative with respect to time, and substitute.
7. Newton's Method for approximating zeros of functions: Let $f(c) = 0$, where f is differentiable on an open interval containing c . Then, to approximate c , we use the following steps:
- Make an initial estimate x_1 , that is close to c . (A graph is helpful.)
 - Determine a new approximation by $x_{n+1} = x_n - f(x_n)/f'(x_n)$.
 - If $|x_n - x_{n+1}|$ is less than the desired accuracy, let x_{n+1} serve as the final approximation. Otherwise, return to step b and calculate a new approximation.

Motion of Particles

1. Rectilinear motion (motion along a straight line)

If $s = f(t)$, then $v_{av} = \Delta s/\Delta t$ = (total distance traveled) / (time elapsed) and $a_{av} = \Delta v/\Delta t$.

- a. $v = ds/dt = f'(t)$. If $v > 0$, then the particle is moving in the direction of increasing s . If $v < 0$, then the particle is moving in direction of decreasing s . If $v = 0$, then the particle is instantaneously at rest. At the instant the particle reverses direction, v must be 0, but not conversely.

- b. $a = dv/dt = f''(t)$. The particle changes direction if $v = 0$ and $a \neq 0$ or v changes sign as we go through $v = 0$. The speed of the particle, $|v|$ (magnitude of v), increases when $a > 0$ and $v > 0$ or $a < 0$ and $v < 0$. The speed decreases when $a < 0$ and $v > 0$ or $a > 0$ and $v < 0$.

- 2.* Curvilinear motion (motion along a curve)

Parametric equations of motion: $x = f(t)$, $y = g(t)$; $v_x = \frac{dx}{dt}$, $v_y = \frac{dy}{dt}$, $a_x = \frac{dv_x}{dt}$, $a_y = \frac{dv_y}{dt}$.

$\|v\| = \sqrt{v_x^2 + v_y^2}$ (magnitude), $\tan \theta = \frac{v_y}{v_x}$ (quadrant depends on signs of v_x and v_y).

$\|a\| = \sqrt{a_x^2 + a_y^2}$ (magnitude), $\tan \alpha = \frac{a_y}{a_x}$ (quadrant depends on signs of a_x and a_y).

Since $\tan \theta = \frac{\frac{dx}{dt}}{\frac{dy}{dt}} = \frac{dx}{dy}$ and $v^2 = v_x^2 + v_y^2 = \left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$, $\|v\| = \left|\frac{ds}{dt}\right|$.

It follows that the velocity vector has is tangent to the curve and the speed of the particle $= \left|\frac{ds}{dt}\right|$, where s is the distance along the curve from some fixed point.

Integration

1. Def'n: Indefinite Integral (Antiderivative) $\int f(x) dx = F(x) + C$ if $F'(x) = f(x)$.

Fundamental Rules:

- $\int (u + v) dx = \int u dx + \int v dx$.
- $\int ku dx = k \int u dx$.
- $\int u^n du = u^{n+1} / (n+1) + C$ ($n \neq -1$).

2. Def'n: Definite Integral

Properties:

- $\int_a^b f(x) dx = -\int_b^a f(x) dx$.
- $\int_a^a f(x) dx = 0$.
- $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$ (for any c).
- $\int_a^b k dx = k(b - a)$.
- $\int_a^b (u + v) dx = \int_a^b u dx + \int_a^b v dx$.

3. Fundamental Theorem of Integral Calculus (Connects concept of indefinite and definite integrals)

If $f(x)$ is continuous on (a, b) , $\int f(x) dx = F(x)$ for any particular integral, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Integration Theorems

- If $f(x)$ is continuous on $[a, b]$, then it is integrable on $[a, b]$. (This works even for finite numbers of discontinuities.)
- If f and g are integrable on $[a, b]$ and $f(x) \leq g(x)$ for all x on $[a, b]$, then $\int_a^b f(x) dx \leq \int_a^b g(x) dx$.
- If f is integrable on $[a, b]$ and $m \leq f(x) \leq M$ for all x in $[a, b]$, then $m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$.
- If $F(x)$ and $G(x)$ are antiderivatives of the same function (i.e. $dF/dx = dG/dx$), then $F(x) = G(x) + C$.
- Average Value of a Function (with respect to x on $[a, b]$): $y_{av} = \frac{1}{b - a} \int_a^b f(x) dx$.

6. Let f be continuous on $[a, b]$. Let x be a number on (a, b) and define $F(x) = \int_a^x f(t) dt$, then F is differentiable and $F'(x) = f(x)$.

7. Change on Variable in Define Integrals: $x = \varphi(t)$, $a = \varphi(t_1)$, $b = \varphi(t_2)$, then $\int_a^b f(x) dx =$

$$\int_{t_1}^{t_2} f(\varphi(t)) \cdot \varphi'(t) dt.$$

Approximations of Integrals

1. Trapezoidal Rule: $\int_a^b y dx \approx \frac{h}{2} (y_0 + 2y_1 + 2y_2 + \dots + 2y_{n-1} + y_n)$, $h = \frac{b-a}{n}$.

$$x_0 = a \quad y_0 = f(x_0) = f(a)$$

$$x_1 = a + h \quad y_1 = f(x_1)$$

$$x_2 = a + 2h \quad y_2 = f(x_2)$$

$$\vdots$$

$$x_n = a + nh \quad y_n = f(x_n) = f(b)$$

$$\epsilon_T = \frac{-(b-a)h^2 f''(c)}{12}, \quad a \leq c \leq b.$$

2. * Area under parabolic arc: $A = (h/3)(y_0 + 4y_1 + y_2)$, where y_0 , y_1 , and y_2 are three equidistant ordinates, and $h = y_1 - y_0 = y_2 - y_1$.

3. * Simpson's Rule: $y = f(x) \geq 0$ on $[a, b]$.

$$\int_a^b y dx \approx \frac{h}{3} (y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + \dots + 4y_{n-1} + y_n), \quad h = \frac{b-a}{n}, \quad n \text{ must be even.}$$

$$\epsilon_s = \frac{-(b-a)h^4 f^{(4)}(c)}{180}, \quad a \leq c \leq b..$$

Basic Integration Formulas

$$1. \int c f(x) dx = c \int f(x) dx.$$

$$3. \int u^n du = u^{n+1}/(n+1) + C \quad (n \neq -1).$$

$$5. \int e^u du = e^u + C.$$

$$7. \int \cos u du = \sin u + C.$$

$$9. \int \tan u du = \ln |\sec u| + C = -\ln |\cos u| + C.$$

$$11. \int \sec^2 u du = \tan u + C.$$

$$13. \int \sec u \tan u du = \sec u + C.$$

$$15. \int \sec u du = \ln |\sec u + \tan u| + C.$$

$$2. \int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx.$$

$$4. \int (1/u) du = \ln |u| + C.$$

$$6. \int a^u du = a^u / \ln a + C \quad (a > 0, a \neq 1).$$

$$8. \int \sin u du = -\cos u + C.$$

$$10. \int \cot u du = \ln |\sin u| + C = -\ln |\csc u| + C.$$

$$12. \int \csc^2 u du = -\cot u + C.$$

$$14. \int \csc u \cot u du = -\csc u + C.$$

$$16. \int \csc u du = \ln |\csc u - \cot u| + C.$$

$$17. \int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin \frac{u}{a} + C.$$

$$18. \int \frac{du}{a^2 + u^2} = \frac{1}{a} \arctan \frac{u}{a} + C.$$

$$19. \int \frac{du}{|u|\sqrt{u^2 - a^2}} = \frac{1}{a} \operatorname{arcsec} \frac{u}{a} + C.$$

Techniques of Integration

1. Substitution: Use of identities and change of variable.
2. Integration by Parts: $\int u dv = uv - \int v du$. (From integrating the product rule: $\int d(uv) = \int u dv + \int v du$.)

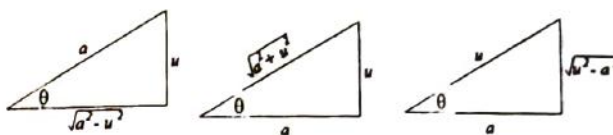
Also Note: For definite integrals: $\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$. Can be used for: $\int x e^x dx$, twice for $\int x^2 \cos x dx$, twice for $\int e^x \sin x dx$, $\int \ln x dx$, $\int \arcsin x dx$.

3. * Trigonometric Substitution:

$u = a \sin \theta$ in $\sqrt{a^2 - u^2}$ yields $a \cos \theta$.

$u = a \tan \theta$ in $\sqrt{a^2 + u^2}$ yields $a \sec \theta$.

$u = a \sec \theta$ in $\sqrt{u^2 - a^2}$ yields $a \tan \theta$.



4. * Partial Fractions: If the original denominator has

a. * A linear factor, place $ax + b$ in the new denominator, with a constant numerator: $\frac{A}{ax + b}$.

b. * A quadratic factor, place $ax^2 + bx + c$ in the new denominator, with a linear numerator:

$$\frac{Ax + B}{ax^2 + bx + c}$$

c. * An unfactorable polynomial to the r power, place:

$$\frac{A_{n-1}x^{n-1} + A_{n-2}x^{n-2} + \dots + A_0x^0}{(a_nx^n + a_{n-1}x^{n-1} + \dots + a_0x^0)^r} + \frac{B_{n-1}x^{n-1} + B_{n-2}x^{n-2} + \dots + B_0x^0}{(a_nx^n + a_{n-1}x^{n-1} + \dots + a_0x^0)^{r-1}} + \dots$$

(until the denominator is raised to the first power).

d. * Any combinations/repetitions of the above cases should be repeated.

5. For $\int \frac{p(x)}{q(x)}$, divide the numerator by the denominator if the former is of higher or of equal degree to the latter.

Completing the Square: Use this technique for quadratic functions if necessary.

Splitting up Fractions: $\int \frac{Ax + B}{ax^2 + bx + c} = \int \frac{Ax}{ax^2 + bx + c} + \int \frac{B}{ax^2 + bx + c}$.

Applications Of The Definite Integral

1. Areas:

a. Area bounded by $y = f(x)$, x -axis, lines $x = a$ and $x = b$ ($b > a$), $A = \int_a^b f(x) dx = \int_a^b y dx$.

b. Area bounded by $x = g(y)$, y -axis, lines $y = a$ and $y = b$ ($b > a$), $A = \int_a^b g(y) dy = \int_a^b x dy$.

c. Using area of a circle $= \pi r^2$, we find $\int_0^a \sqrt{a^2 - x^2} dx = \frac{1}{4} \pi a^2$.

d. If $f(x) \geq g(x)$ on $[a, b]$, then the area bounded by the curves $y_1 = f(x)$, $y_2 = g(x)$ and the lines $x = a$

and $x = b$, $A = \int_a^b (y_2 - y_1) dx = \int_a^b (g(x) - f(x)) dx$.

c. * Polar Coordinates:

i. * The area bounded by $r = f(\theta)$ and two radii vectors $\theta = \alpha$ and $\theta = \beta$ is $A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$.

ii. * The area between $r_1 = f(\theta)$ and $r_2 = g(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is $A = \frac{1}{2} \int_{\alpha}^{\beta} (r_1^2 - r_2^2) d\theta$.

2. Volumes:

a. **Known Cross-Sectional Area:** Volume of solid with cross-sectional area $A(x)$ perpendicular to x -

axis at x is: $V = \lim_{n \rightarrow \infty} \sum_{k=1}^n A(c_k) \Delta x_k = \int_a^b A(x) dx$.

b. **Solid of Revolution:** A solid generated by revolving a planar region about an axis in its plane (plane perpendicular to axis cuts circular cross sections).

c. Volume of solid formed by revolving region bounded by $y = f(x)$, $x = a$, $x = b$ ($b > a$), about x -axis

is $V = \pi \int_a^b y^2 dx$ (Cylindrical Disks).

d. Volume of solid formed by revolving region bounded by $x = g(y)$, $y = a$, $y = b$ ($b > a$), about y -axis

is $V = \pi \int_a^b x^2 dy$.

e. If $f(x) \geq g(x)$ on $[a, b]$, volume generated by revolving about x -axis region bounded by curves $y_1 =$

$f(x)$, $y_2 = g(x)$, $x = a$ and $x = b$ ($b > a$) is $V = \pi \int_a^b (y_2^2 - y_1^2) dx$ (cylindrical washers).

f. Volume of solid generated by revolving region bounded by $y = f(x)$, x -axis, $x = a$, $x = b$ ($b > a$)

about y -axis is $V = 2\pi \int_a^b xy dx$ (cylindrical shells). Volume of solid generated by revolving region

bounded by $x = g(y)$, y -axis, $y = a$, $y = b$ about x -axis $V = 2\pi \int_a^b yx dy$.

3. * Arc Length:

a. * If $y = f(x)$, and $y' = f'(x)$ are continuous, then the arc length of $y = f(x)$ from $x = a$ to $x = b$ ($b > a$) is

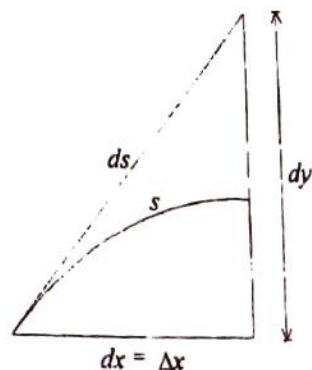
$s = \int_a^b \sqrt{1 + y'^2} dx$. Similarly for $x = g(y)$, $x' = g'(y)$ from $y = a$ to $y = b$, $s = \int_a^b \sqrt{1 + x'^2} dy$.

b. * The differential of arc for a smooth curve $y = f(x)$, $ds =$

$\sqrt{1 + y'^2} dx$. Squaring, we get $(ds)^2 = (dx)^2 + (dy)^2$ and the above formula can be written as $s = \int ds$. Also note the geometric interpretation.

c. * Parametrically, if $x = f(t)$, $y = g(t)$, then since $(ds)^2 = (dx)^2 + (dy)^2$, taking the square root and integrating yields $s =$

$\int_{t_0}^{t_1} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$.



d.* Polar Coordinates: Length of curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is $s = \int_{\alpha}^{\beta} \sqrt{r^2 + r'^2} d\theta$.

4.* Work: Motion Along Straight Line

a.* $W = fd$ (for a constant force only).

b.* Total work done in moving an object along x -axis from a to b , if $f(x)$ is a force exerted at a point x ,

$$\text{is } W = \int_a^b f(x) dx.$$

c.* Hooke's Law: $f \propto x$ or $f = kx$. (k is the spring constant, i.e. the force required to stretch the spring one unit from equilibrium position.)

5.* Improper Integrals

a.* Convergent: approaches a limiting value. Divergent: Approaches no value (∞ , oscillates, etc.).

b.* Type 1 (Discontinuous Integrand)

i.* Integrand becomes ∞ at lower limit: $\int_a^b f(x) dx = \lim_{h \rightarrow 0^+} \int_{a+h}^b f(x) dx.$

ii.* Integrand becomes ∞ at upper limit: $\int_a^b f(x) dx = \lim_{h \rightarrow 0^+} \int_a^{b-h} f(x) dx.$

iii.* c ($a < c < b$) makes integrand ∞ : $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, use i and ii.

c.* Type 2 (Infinite Limits)

i.* $\int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$

ii.* $\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx.$

iii.* $\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx$ (for suitable c).

More on Limits

1. Squeeze Theorem: If $A \leq f(x) \leq g(x)$ for some constant A and $\lim_{x \rightarrow c} g(x) = A$, then $\lim_{x \rightarrow c} f(x) = A$.

2. Important Limits:

a. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$

b. $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1.$

c. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0.$

d. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1.$

e. $\lim_{h \rightarrow \infty} \left(1 + \frac{1}{h}\right)^h = \lim_{h \rightarrow 0} (1 + h)^{\frac{1}{h}} = e.$

f. $\lim_{k \rightarrow \infty} \left(1 + \frac{n}{k}\right)^k = \lim_{\left(\frac{k}{n} \rightarrow \infty\right)} \left[\left(1 + \frac{1}{\frac{k}{n}}\right)^{\frac{k}{n}}\right]^n = e^n.$

* Note: The rest of the material applies only to the BC curriculum.*

Sequences and Series

1. Definitions

- Def'n: A sequence is a function whose domain is the set of positive integers. (The numbers in the range of the sequence are its elements or terms.)
- Def'n: A sequence $\{a_n\}$ has the limit L , written $\lim_{n \rightarrow \infty} a_n = L$, if to each number $\epsilon > 0$, there corresponds an integer $N > 0$ such that $|a_n - L| < \epsilon$ whenever $n \geq N$ (i.e. for any $\epsilon > 0$ there exists an integer $N > 0$ such that all terms beyond the N th in the sequence differ from L by less than ϵ).
- Def'n: A sequence that has a limit is said to be convergent, and to converge to that limit. A sequence that fails to have a limit is divergent.

2. Basic Theorems

- Thm: If $\lim_{x \rightarrow \infty} f(x) = L$ and f is defined for all positive integers, then also $\lim_{n \rightarrow \infty} f(n) = L$, where n is a positive integer.
- Thm: The limit of a convergent sequence is unique.
- Thm: If $\{a_n\}$ and $\{b_n\}$ are convergent sequences and C is a constant, then
 - $\lim_{n \rightarrow \infty} ca_n = c \lim_{n \rightarrow \infty} a_n$.
 - $\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$.
 - $\lim_{n \rightarrow \infty} a_n b_n = \lim_{n \rightarrow \infty} a_n \lim_{n \rightarrow \infty} b_n$.
 - $\lim_{n \rightarrow \infty} (a_n/b_n) = \lim_{n \rightarrow \infty} a_n / \lim_{n \rightarrow \infty} b_n$ (provided $\lim_{n \rightarrow \infty} b_n \neq 0$).

3. Monotonic Sequences

- Def'n: A sequence is said to be monotonic and
 - Increasing (or non-decreasing) if $a_n \leq a_{n+1}$ (i.e. $a_1 \leq a_2 \leq a_3 \leq \dots$).
 - Strictly increasing if $a_n < a_{n+1}$ (i.e. $a_1 < a_2 < a_3 < \dots$).
 - Decreasing (or non-increasing) if $a_n \geq a_{n+1}$ (i.e. $a_1 \geq a_2 \geq a_3 \geq \dots$).
 - Strictly decreasing if $a_n > a_{n+1}$ (i.e. $a_1 > a_2 > a_3 > \dots$).

4. Bounds

- Definitions
 - " A " is called an upper bound of the sequence $\{a_n\}$ if $a_n \leq A$ for all positive integers n .
 - " B " is called a lower bound of the sequence $\{a_n\}$ if $a_n \geq B$ for all positive integers n .
 - A sequence $\{a_n\}$ is said to be bounded if there is a number " C " such that $|a_n| \leq C$ for all positive integers n . (Equivalently, $\{a_n\}$ is bounded if and only if it has an upper bound and a lower bound.)
- Axiom of completeness: Every set of real numbers which has a lower bound has a greatest lower bound (glb), and every set of real numbers which has an upper bound has a least upper bound (lub).
- Thm: A monotonic sequence is convergent if and only if it is bounded. (Note: For all sequences $\{a_n\}$ convergence implies boundedness, but not conversely.)
- More Theorems
 - If $\{a_n\}$ is a monotonically increasing sequence which has an upper bound, then the sequence is convergent, its limit being the lub of the numbers $\{a_n\}$.

- ii. If $\{a_n\}$ is a monotonically decreasing sequence which has a lower bound then the sequence is convergent, its limit being the *g/lb* of the numbers $\{a_n\}$.

Infinite Series

1. Definitions and Basic Theorems

- a. Def'n: If $\{a_n\}$ is a sequence, then an expression of the form $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \dots + a_n + \dots$ is called an infinite series.
- b. Def'n: Let $\sum_{n=1}^{\infty} a_n$ be an infinite series, and let $\{S_n\}$ where $S_n = a_1 + a_2 + a_3 + \dots + a_n$ be the sequence of the partial sums of the series. If $\lim_{n \rightarrow \infty} S_n$ exists and is equal to S , the series is said to be convergent and S is called the sum of the infinite series $\sum_{n=1}^{\infty} a_n$. If $\lim_{n \rightarrow \infty} S_n$ fails to exist, the series is divergent and has no sum (If series $\sum_{n=1}^{\infty} a_n$ has a sum, we write $S = \sum_{n=1}^{\infty} a_n$).
- c. Thm: (necessary condition for convergence) If the infinite series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$. (Note: converse is not true.)
- d. Corollary: If the n th term of an infinite series $\sum_{n=1}^{\infty} a_n$ does not approach 0 as $n \rightarrow \infty$, the series is divergent.
- e. Thm: The two infinite series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=k}^{\infty} a_n$, for any positive integral k , both either converge or diverge.
- f. Thm: If " c " is any non-zero constant and $\sum a_n$ converges, then so does $\sum ca_n$, and its limit is $c \sum a_n$. If $\sum a_n$ diverges, then so does $\sum ca_n$.
- g. Thm: If $\sum a_n$ and $\sum b_n$ are convergent series, then $\sum (a_n + b_n)$ and $\sum (a_n - b_n)$ are also convergent, and $\sum (a_n \pm b_n) = \sum a_n \pm \sum b_n$.
- h. Thm: The terms of a convergent series may be grouped in any way and the new series will converge to the same limit as the original series.
- i. Thm: let $\{S_n\}$ be the sequence of partial sums of $\sum a_n$, then for every $\epsilon > 0$ there exists a number N such that $|S_n - S_m| < \epsilon$ whenever $n, m > N$.
2. Some Important Series
- a. The Geometric Series: $\sum r^n$ is convergent for $|r| < 1$ with sum $= a/(1 - r)$ and divergent for all other values of r .
- b. The Harmonic Series: $\sum 1/n$ is divergent.

- c. The Hyperharmonic series $\sum 1/(n^k)$ is convergent for $k > 1$, and divergent for $k \leq 1$.
 - d. The Telescopic Series $\sum 1/(n(n+1))$ is convergent.
 - e. The series $\sum 1/n!$ is convergent.
 - f. Polynomial Test: If $P(n)$ and $Q(n)$ are polynomials of p th and q th degree respectively, then $\sum P(n)/Q(n)$ converges if $q > p + 1$ and diverges if $q \leq p + 1$.
3. Tests For Convergence of Series of Positive Term
- a. Defn: Series of positive terms only will be called positive series.
 - b. Thm: A necessary and sufficient condition for an infinite positive series to be convergent is that its sequence of partial sums has an upper bound.
 - c. Thm: (Comparison Test) Let $\sum a_n, \sum b_n, \sum c_n$ be positive series.
 - i. If $\sum b_n$ is convergent and $a_n \leq b_n$ (for all n), then $\sum a_n$ is also convergent.
 - ii. If $\sum c_n$ is divergent and $a_n \geq c_n$ (for all n), then $\sum a_n$ is also divergent.

d. Thm: (Integral Test) If f is positive, continuous, and decreasing for $x \geq 1$ and $a_n = f(n)$, then $\sum_{n=1}^{\infty} a_n$

and $\int_1^{\infty} f(x) dx$ either both converge or both diverge. If the series converges to S , then the remainder

$$R_N = S - S_N \text{ is bounded by } 0 \leq R_N \leq \int_N^{\infty} f(x) dx.$$

e. Thm: (Ratio Test) let $\sum a_n$ be a positive series, and let $\lim_{n \rightarrow \infty} (a_{n+1})/a_n = L$.

- i. If $L < 1$, the series converges.
- ii. If $L > 1$, the series diverges.
- iii. If $L = 1$, the test fails.

4. Alternating Series and Absolute Convergence

a. Defn: An infinite series of the form $\sum_{k=1}^{\infty} (-1)^{k+1} (a_k) = a_1 - a_2 + a_3 - \dots$ where $a_k > 0$ (for all k), is called an alternating series.

b. Thm: The alternating series $a_1 - a_2 + a_3 - \dots$ converges if $0 < a_{n+1} < a_n$ (for all n) and $\lim_{n \rightarrow \infty} a_n = 0$.

c. Thm: If $\sum_{k=1}^{\infty} (-1)^{k+1} (a_k) = a_1 - a_2 + a_3 - \dots$ is an alternating series such that $0 < a_{k+1} < a_k$ and $\lim_{k \rightarrow \infty} a_k = 0$, the error made by using the sum of the first k terms as an approximation for the sum of the series is less than a_{k+1} . The absolute value of the first neglected term, i.e.

$$\left| \sum_{n=1}^{\infty} (-1)^{n+1} a_n - \sum_{n=1}^k (-1)^{n+1} a_n \right| < a_{k+1}$$

d. Defn:

- 1. The infinite series $\sum a_n$ converges absolutely if $\sum |a_n|$ converges.
- 2. If a series is convergent but not absolutely convergent, it is said to be conditionally convergent.

e. Thm: If a series converges absolutely, then it also converges.

f. Thm: For any series of nonzero terms $\sum a_n$, let $\lim_{n \rightarrow \infty} |(a_{n+1})/a_n| = L$.

1. If $L < 1$, series converges absolutely.
2. If $L > 1$, series diverges.
- g. Thm: If a series is absolutely convergent, its terms may be rearranged without affecting the absolute convergence of the series or its sum.
- h. Thm: If $\sum a_n$ and $\sum b_n$ are absolutely convergent series with sums A and B , respectively, then the series $a_1b_1 + (a_1b_2 + a_2b_1) + (a_1b_3 + a_2b_2 + a_3b_1) + \dots$ converges absolutely and its sum is AB .

Power Series

1. Definitions and Basic Theorems

- a. Defn: If $\{a_n\}$ is a sequence of constants, the expression $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$ is called a power series in x .
- b. If the power series $\sum a_n x^n$ converges for a number $x_1 \neq 0$ then it converges absolutely for all number x such that $|x| < |x_1|$.
- c. If the power series diverges for a number x_2 , then it diverges for all numbers x such that $|x| > |x_2|$.
- d. Let $\sum a_n x^n$ be a given power series. Then exactly one of the following conditions holds:
 - i. The series converges only when $x = 0$.
 - ii. The series is absolutely convergent for all values of x .
 - iii. There exist a number $r > 0$ (called the radius of convergence) such that the series is absolutely convergent for all values of x for which $|x| < r$ and is divergent for all values of x for which $|x| > r$ (converges possibly at one or both endpoints $r, -r$).
- e. Defn: The set of all values for which a power series is convergent is called the interval of convergence of the power series.
- f. Similarly, if a_n is a sequence we define the expression $\sum_{n=0}^{\infty} a_n (x - b)^n = a_0 + a_1(x - b) + a_2(x - b)^2 + a_3(x - b)^3 + \dots$ to be a power series in $(x - b)$.
- g. The conditions in the above theorem now become:
 - i. The series converges only when $x = b$.
 - ii. The series is absolutely convergent for all values of x .
 - iii. There exists a number $r > 0$ such that the series is absolutely convergent for all values of x for which $|x - b| < r$ and is divergent for all values of x for which $|x - b| > r$ (converges possibly at one or both endpoints $b - r, b + r$).

2. More on Power Series

- a. Defn: Since the power series $\sum a_n (x - b)^n$ has a unique sum at each point in its interval of convergence, a function f is defined by the power series, i.e. $f(x) = \sum a_n (x - b)^n$. The domain of f is the interval of convergence of the power series, and the values of f at any point x in this domain is the sum of the power series at this point.
- b. Defn: The derived series of the power series $\sum a_n (x - b)^n$ is $\sum n a_n (x - b)^{n-1} = a_1 + 2a_2(x - b) + 3a_3(x - b)^2 + \dots + n a_n (x - b)^{n-1} + \dots$, i.e., the series formed by taking the derivatives of the terms of the original series.
- c. Functions defined by power series behave very much like polynomials as is indicated by the following theorem. Thm: Let $\sum a_n (x - b)^n$ be a power series in $(x - b)$ and let f be defined by $f(x) = \sum a_n (x - b)^n$, where x is any number in the interval of convergence of the series.
 - i. f is continuous at every point in the interval of convergence.

ii. The derived series of the power series is equal to the derivative of the function defined by the power series at every point in the interval of convergence, except possibly at the endpoints, i.e. $f(x) = \sum D_x [a_n(x-b)^n] = \sum na_n(x-b)^{n-1}$.

iii. The series obtained by integrating the terms of the given series converge to $\int_b^x f(t) dt$ for each x within the interval of convergence of the original power series, except possibly at the

$$\text{endpoints, i.e. } \int_b^x f(t) dt = \sum_{n=0}^{\infty} \int_b^x a_n(t-b)^n dt = \sum_{n=0}^{\infty} a_n \frac{(x-b)^{n+1}}{n+1} = a_0(x-b) + a_1(x-b)^2 + \dots$$

(Note: It follows from ii above that a function that can be represented by a power series is infinitely differentiable and all derived series have the same radius of convergence (not necessarily same interval).)

3. Taylor Series and Taylor's Theorem

a. If a function f is infinitely differentiable, the series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(b)(x-b)^n}{n!} = f(b) + f'(b)(x-b) + \frac{f''(b)(x-b)^2}{2!} + \frac{f'''(b)(x-b)^3}{3!} + \dots$$

is called the

Taylor series in $(x-b)$ of the function.

b. Consider any power series in $(x-b)$, $a_0 + a_1(x-b) + a_2(x-b)^2 + \dots$ and denote its interval of convergence by $(b-r, b+r)$, where $0 \leq r < \infty$. Denote by the function defined by $f(x) = a_0 + a_1(x-b) + a_2(x-b)^2 + \dots + a_n(x-b)^n + \dots$ where x is any number in $(b-r, b+r)$. Then the power series is the Taylor's series in $(x-b)$ of the function f , namely $f(x) = f(b) + f'(b)(x-b) + f''(b)(x-b)^2 / 2! + \dots + f^{(n)}(b)(x-b)^n / n! + \dots$ where $b-r < x < b+r$ (i.e. $a_n = f^{(n)}(b) / n!$).

c. The Taylor series in x is called a Maclaurin series, $f(0) + f'(0)x + f''(0)x^2 / 2! + f'''(0)x^3 / 3! + \dots + f^{(n)}(0)x^n / n! + \dots$. A Taylor series in $(x-b)$ of a function f is sometimes called the Taylor expansion of f about the point b .

d. Taylor's Theorem: Let f be a function which is defined in some open interval $(b-r, b+r)$, $0 \leq r < \infty$, and whose first $n+1$ derivatives exist and are continuous throughout the interval. Then for all numbers x in $(b-r, b+r)$, $f(x) = f(b) + f'(b)(x-b) + f''(b)(x-b)^2 / 2! + \dots + f^{(n)}(b)(x-b)^n / n! + R_n(x)$ where $R_n(x) = f^{(n+1)}(\xi)(x-b)^{n+1} / (n+1)!$ (Lagrange form of the remainder) where $b < \xi < x$.

e. Other forms of the remainder are:

i. Integral form: $R_n(x) = 1/n! \int_b^x (x-t)^n f^{(n+1)}(t) dt.$

ii. Cauchy's form: $R_n(x) = f^{(n+1)}(\xi)(x-\xi)^n(x-b) / n!$, where ξ is some number between b and x .

f. Thm: Since $f(x) - \sum_{k=0}^n f^{(k)}(b)(x-b)^k / k! = R_n(x)$, it follows that if f is a function having derivatives of all orders in some interval $(b-r, b+r)$. A necessary and sufficient condition that the Taylor's

series $\sum_{n=0}^{\infty} f^{(n)}(b)(x-b)^n / n!$ represents the function f at all points in the interval is that, whenever $|x$

$-b| < r$, $\lim_{n \rightarrow \infty} R_n(x) = 0$, where $R_n(x)$ is the remainder after $n+1$ terms in Taylor's formula.

Important Series Expansions

$$1. \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^{n-1} x^{2n-1}}{(2n-1)!} + \dots \quad -\infty < x < \infty.$$

$$2. \sin x = \sin a + (x-a)\cos a - \frac{(x-a)^2 \cos a}{2!} - \frac{(x-a)^3 \cos a}{3!} + \dots \quad -\infty < x < \infty.$$

$$3. \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^{n-1} x^{2n-2}}{(2n-2)!} + \dots \quad -\infty < x < \infty.$$

$$4. \cos x = \cos a - (x-a)\sin a - \frac{(x-a)^2 \cos a}{2!} + \frac{(x-a)^3 \sin a}{3!} + \dots \quad -\infty < x < \infty.$$

$$5. e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots \quad -\infty < x < \infty.$$

$$6. \frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots \quad -1 < x < 1.$$

$$7. \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad -1 \leq x < 1.$$

$$8. \frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots \quad -1 < x < 1.$$

$$9. \ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots \quad -1 < x < 1.$$

$$10. \ln(x) = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots \quad 0 < x \leq 2.$$

$$11. \frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots \quad -1 < x < 1.$$

$$12. \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \quad -1 \leq x < 1.$$

$$13. \text{The General Binomial Theorem: (for all real } m, -1 < x < 1)$$

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!} x^2 + \dots + \frac{m(m-1)\dots(m-n+1)}{n!} x^n + \dots$$

Daniel Russo

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